

Polluted Beginnings: Air Pollution and Child Health

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Abstract

Bangladesh consistently ranks among the most polluted countries in the world and continues to face significant challenges in child health. Using data from Bangladesh Demographic and Health Survey (BDHS), this study examines the impact of cumulative lifetime exposure to particulate matter on multiple child growth indicators. I exploit the exogenous occurrence of thermal inversions to capture quasi-random variation in particulate matter exposure. I find that a 1 $\mu\text{g}/\text{m}^3$ increase in pollution reduces average height for age (HAZ) score by 0.014 standard deviations and increases the probability of stunting by 0.48 percentage points. By contrast, the effects on weight-based measures—weight-for-height, weight-for-age, severe stunting, and underweight—are not consistently statistically significant. Back-of-the-envelope calculations indicate that a one standard deviation increase in lifetime exposure translates into a loss of 0.5–0.8 cm in adult height and a 0.4–0.8% reduction in lifetime earnings, with women facing larger earnings penalties than men despite smaller height losses, owing to a higher height premium.

Keywords: pollution, child health, anthropometry

JEL classifications: C36, I15, Q53

1 Introduction

In recent years, air pollution has emerged as a major environmental concern, ranking second in the list of global risk factors for death after high blood pressure. In 2021, nearly 8.1 million deaths -or 1 in 8 deaths worldwide-were attributable to air pollution, of which 58% were due to PM_{2.5}, 38% from household air pollution and 6% from ozone (Health Effects Institute 2024). The economic burden was also considerable: the economic costs associated with child growth failure amounted to 16% of global GDP in the same year (Xu, Feng and Shao 2024). South Asia bears a disproportionate share of this burden, with Bangladesh recording 236,000 pollution-related deaths in 2021, the second highest toll in the region after Pakistan (Health Effects Institute 2024).

Prolonged exposure to air pollution worsens health outcomes by leading to increased mortality and morbidity. It increases the incidence of respiratory ailments as well as cardiovascular and respiratory diseases (Aguilar-Gomez et al. 2022). Nearly 30% of deaths from lower respiratory infections, 28% from ischemic heart disease and 48% from chronic obstructive pulmonary disease are attributable to air pollution (Health Effects Institute 2024).

Pollution induced health disparities among children are important to study as early life exposure to pollution can generate persistent impacts over the child's life (Isen et al. 2017; Baran et al. 2026) and also lead to intergenerational transmission of inequality (Currie 2009). Because of having underdeveloped immune systems and lung function, children are particularly susceptible to the adverse impacts of ambient air pollution. Relative to adults, children are more susceptible to the effects of air pollution as they absorb more pollutants and inhale more air per kilogram of body weight. Furthermore, they are also more likely to spend more time outdoors and breathe air closer to the ground which puts them in closer proximity to pollution sources like dust and vehicle exhaust. Because of this, exposure to fine particulate matter early in life is associated with increased susceptibility to respiratory infections such as pneumonia and asthma. It also impairs physical and cognitive development including growth faltering, high blood pressure, childhood leukemia and cognitive deficits

(Health Effects Institute 2024).

Among children, air pollution ranks second in place as a global risk factor for death and is only surpassed by malnutrition (Health Effects Institute 2024). In 2021, nearly 709,000 deaths of children under 5, representing 15% of all global deaths in children under 5, were due to air pollution, with the largest burden of disease being reported in Asia and Africa (Health Effects Institute 2024). According to WHO, nearly 250 million children under 5 years of age in low- and middle-income countries are at risk of adequate development and growth (WHO 2020). This vulnerability is further compounded in developing and low-income countries where limited remediation capacity coupled with weak enforcement of environmental regulations puts children at higher risk and increases their likelihood of suffering from long-term health consequences.

Early life health conditions play a critical role in shaping individuals' socioeconomic trajectories by affecting cognitive development, educational attainment and overall well-being. Because of this, early exposure to air pollution can impose sizable and persistent social and economic costs. Quantifying these impacts is critical for evaluating the benefits of pollution mitigation policies and for determining progress towards the United Nations Sustainable Development Goal of reducing under-5 mortality.

The Bangladeshi context is well suited to examine the relation between ambient air pollution and child health outcomes for two primary reasons. First, the impact of pollution on neonatal outcomes is worse in middle- and low-income countries (LMICs). The average annual pollution levels in Bangladesh are as high as $79.9 \mu\text{g}/\text{m}^3$ as compared to the United States of $10 \mu\text{g}/\text{m}^3$ (Misra and Kulshreshtha 2025). Second, baseline maternal health may be poorer in developing countries due to limited access to healthcare, undernutrition or higher disease burden, further exacerbating fetal growth and development.

A vast body of literature looks at the impact of pollution on child health (Chay and Greenstone 2003, Chen et al. 2013; Arceo, Hanna and Oliva 2016; Knittel, Miller and Sanders 2016, Currie and Walker 2011, Simenova et al. 2019). Most of the literature provides

evidence from the developed world (Janke 2014, Hill 2018, Sinharoy et al. 2020, Klauber et al. 2024). Evidence from developing countries remains limited, with most papers being published recently (Balietti, Datta and Veljanoska 2022, Zhang et al. 2020). I focus on the impact of pollution on child health both because of its importance over the lifecycle and overall well-being and because pollution affects these outcomes over short time horizons (Alexander and Schwandt 2019; Li and Zhang 2024). Prior literature has examined the impact of pollution on child growth indicators by varying the duration and intensity of pollution exposure. The effects of in utero exposure to pollution have been extensively explored (Adjei-Mantey and Takeuchi 2021) in the first (Michikawa et al. 2017; Pereira et al. 2014), second (Hannam et al. 2014; Wang et al. 2018) and third trimester (DeFranco et al. 2016; Rich et al. 2015). This paper contributes to the prevalent environmental health literature by causally estimating the impact of pollution on child growth outcomes in Bangladesh.

Identifying the causal effect of air pollution requires addressing the empirical challenge of endogeneity. Pollution exposure may be correlated with factors that independently affect child health such as residential sorting, socioeconomic status and local economic conditions. Therefore, I use thermal inversions as an instrument to evaluate child health effects. Thermal inversions trap pollutants near the ground and cause increases in air pollution. I instrument lifetime pollution exposure with lifetime exposure to thermal inversions from conception to date of interview. Results from my first stage regression reveal that a one standard deviation increase in thermal inversions is associated with a $4.5 \mu\text{g}/\text{m}^3$ increase in PM_{2.5} levels. My baseline specification finds that a $1 \mu\text{g}/\text{m}^3$ increase in average pollution levels reduces average height for age (HAZ) score by 0.014 standard deviations and increases the probability of stunting by 0.48 percentage points. The estimate for severe stunting is positive but not statistically significant. Weight for height (WHZ) is positive and significant (and wasting is negative and marginally significant). The weight-for-age (WAZ), and underweight estimates are not statistically distinguishable from zero. These results indicate that lifetime pollution

exposure primarily impairs height-based growth rather than acute, weight-based outcomes. To interpret the magnitude of these estimates, I quantify the long-run economic costs of early-life pollution exposure. Following Baliatti, Datta and Veljanoska (2022), I translate the estimated reduction in child height into a loss in adult height using age- and gender-specific correlations between childhood and adult height (Molinari et al. 1995), and then into forgone earnings using the height premium estimated for Indonesia by Sohn (2015). A one standard deviation increase in lifetime pollution exposure ($10.5 \mu\text{g}/\text{m}^3$) implies a reduction in adult height of 0.58–0.80 cm for men and 0.47–0.61 cm for women, and a corresponding fall in lifetime earnings of 0.43–0.60% for men and 0.60–0.79% for women. Although women experience smaller absolute height losses, they face larger earnings penalties, since the height premium is substantially higher for women (13% per 10 cm) than for men (7.5%).

I employ various robustness checks to ensure my results hold across alternative specifications. My checks include recreating the thermal inversion variable using alternate temperature layers, using thermal inversion strength to determine the severity of thermal inversions, controlling flexibly for meteorological controls, excluding meteorological controls, winsorizing child health indicators, including division and wave fixed effects instead of cluster and birth cohort fixed effects and restricting sample to gestational window and the first 1000 days only. I combine household health and anthropometric data from the Bangladesh Demographic and Health Survey (BDHS) with high resolution satellite data for pollution from European Center for Medium-Range Weather Forecasts (ECMWF) and thermal inversions from Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA2). I aggregate pollution concentrations and thermal inversion counts for each child in the sample from gestational period till the time of the interview. I utilize the exogenous variation in thermal inversions to instrument for pollution and determine their impact on child growth outcomes.

This paper contributes to the existing literature on air pollution and child health. It investigates the impact of lifetime pollution exposure on child growth indicators in Bangladesh, exploiting thermal inversions as a source of exogenous variation in pollution. I utilize high

resolution satellite data to determine the impact of air pollution on child growth indicators by focusing on variation in medium-term pollution and determining the life-time exposure of pollution since conception. This means that the time period from conception to the age of child at the time of interview is covered (Alexander and Schwandt 2019, Balietti, Datta and Veljanoska 2022). Findings from this analysis aim to deepen understanding of how air pollution critically influences children’s nutritional well-being and health in developing countries.

The remainder of the paper is organized as follows: section 2 describes the context while section 3 describes the data and descriptive statistics used in the paper. Section 4 presents the empirical analysis including main results, heterogeneity and robustness checks. Section 5 concludes the paper.

2 Context

2.1 Air Pollution in Bangladesh

Air pollution poses a severe public health risk in Bangladesh. In 2023, Bangladesh ranked as the most polluted country globally, with annual average pollution levels of $79.9 \mu\text{g}/\text{m}^3$ -more than twice the national standard of $35 \mu\text{g}/\text{m}^3$ and 15 times the World Health Organization’s (WHO) threshold of $5 \mu\text{g}/\text{m}^3$ (Nesan, Thieriot and Kelly 2025). Annually, pollution exposure leads to 102,456 deaths, of which 5,258 deaths are of children under 5 (Nesan, Thieriot and Kelly 2025). It also leads to 900,485 preterm births and 696,389 low birth weight births annually. The resulting economic burden amounts to \$11 billion annually or nearly 4.4-4.8% of Bangladesh’s GDP (Nesan, Thieriot and Kelly 2025). Among types of pollutants, PM2.5 is of paramount concern. Less than $2.5\mu\text{m}$ in aerodynamic diameter, PM2.5 can remain airborne for long periods and also lodge deep into the lungs (Aguilar-Gomez et al. 2022). They are emitted from various sources including industrial emissions, vehicle exhaust and wildfires. Air pollution impacts respiratory tract following inhalation and leads to lung

inflammation and impedes cellular function in the body. It also impacts the cardiovascular system and leads to changes in blood pressure and heart rate variability.

2.2 Air Pollution and Child Health

There is considerable literature documenting the impact of ambient air pollution on infant health, impaired fetal growth and mortality (Alexander and Schwandt 2022; Arceo, Hanna and Oliva 2016; Currie and Neidell 2005; Currie and Walker 2011; Currie et al. 2015; Knittel, Miller and Sanders 2016; Deryugina et al. 2019). Prior literature has also determined the impact of early life influencing factors on child nutrition (Blom et al. 2022; Shrinivas, Baylis and Crost 2025). Studies have focused on determining both the impact of both short-term (Jan et al. 2018; Kurata et al. 2020) and long-term including lifetime pollution exposure on child growth (Balietti, Datta and Veljanoska 2022; Jayachandran 2009; Currie, Mullins and Neidell 2013; Zhang et al. 2020). Literature has also looked at the impact of indoor pollution on child health (Adjei-Mantey and Takeuchi 2021; Ai and Tan 2025; Wang, Bian and Zhang 2023) and ambient air pollution (Goyal and Canning 2018). While studies on indoor air pollution emphasize on the household’s ability to mitigate exposure through fuel choice, exposure to elevated levels of ambient air pollution is largely unavoidable. Because of this, its impact on child health outcomes warrants greater investigation.

Pollution exposure leads to respiratory infections that can be countered by the immune system. Because their immune system is still developing, children are more adversely impacted by higher pollution concentrations. Prolonged pollution exposure leads to worse respiratory symptoms such as lower respiratory infections and asthma, compromised immune system development and increases the risk of nutritional imbalances among children (Xu et al. 2024). Exposure to ambient air pollution can affect child growth through two channels (Ai and Tan 2025). First, maternal exposure to high pollution concentrations increases the risk of both physical and psychological illnesses including respiratory infections and lung disease. These can have a detrimental impact on fetal development either by di-

rect disease transmission or indirect maternal stress and reduced nutrition intake (Ai and Tan 2025). Maternal stress and hypertension can also impair fetal growth and alter child's early-life sleep patterns, which are closely related to child growth (Ai and Tan 2025). Second, prolonged exposure to pollution can increase the incidence of respiratory illnesses and weaken their immune system, further exacerbating child health indicators.

Child growth indicators include height for age (HAZ) z score, weight for height (WHZ) z score, weight for age (WAZ) z score, stunting, severe stunting, wasting, severe wasting and underweight. Stunting (low height for age), or height for age z score, is an indicator for impaired growth and development and is related to slow skeletal development (Balietti, Datta and Veljanoska 2022). Arising from persistent or recurrent undernutrition, it is defined as a child being too short for their age and is determined with the child's height being more than two standard deviations below the World Health Organization (WHO) Child Growth Standards median. Children whose heights fall below 3 standard deviations are classified as severely stunted. Globally, nearly 162 million children are stunted (Adjei-Mantey and Takeuchi 2021). One of the sustainable development goals (SDGs) is to end all forms of malnutrition by 2030.

Resulting from chronic or recurrent undernutrition, stunting or being underweight is associated with short and long-term physical and socioeconomic disadvantages including reduced cognitive and physical development. It is also associated with weakened immune system and increased susceptibility to other health conditions including pulmonary and other respiratory infections and higher mortality rates (Adjei-Mantey and Takeuchi 2021). In the long term, early life nutritional deprivation is linked to long-run child health and nutrition shocks leading to reduced height in adulthood, worse cognitive development leading to worse education outcomes, lower income, higher obesity risk, higher risk of degenerative diseases such as diabetes and chronic diseases and worse marriage and reproductive outcomes (Hoddinott et al. 2013; Maluccio et al. 2009; Black et al. 2013; Vogl 2014; Xu et al. 2024).

Wasting (low weight for height) is used to assess chronic undernutrition and is linked

to inadequate quality or quantity of food consumption or recurrent or prolonged illnesses. Children whose weight for height falls under two standard deviations are classified as wasted and under 3 deviations as severely wasted.

Underweight (weight for age z score) is when the child’s weight falls below two standard deviations. In 2011, stunting affects nearly 37.87% of children while this percentage drops marginally to 34.06% by 2014 and to 30.31% by 2017-18. Nearly 13.5% of children are severely stunted in 2011 10.45% in 2014 and 8.39% in 2017-18. The percentage of wasted children also decreased from 14.96% in 2011 to 14.30% in 2014 and 8.17% in 2017-18. In 2011, 3.58% of children were severely wasted with this percentage dropping to 3.1% in 2014 and 1.51% in 2017-18.

3 Data and Descriptive Statistics

3.1 Demographic and Health Survey

This paper uses child growth outcomes from Bangladesh Demographic and Health Survey. Nationally representative, these datasets provide information on key metrics of social progress including background characteristics, marriage and sexual activity, fertility, infant, child and maternal health and mortality, nutritional status and female empowerment details. I use 3 rounds of BDHS which were conducted during the years 2011, 2014 and 2017-18 and created a pooled cross-sectional dataset. It includes responses from ever-married women aged 15-49 years regarding children born in the past 5 years since the date of the survey. This refers to children born between the years 2006-18. The data also has cluster coordinates, randomly sampled from census enumeration areas. Regionally stratified, this data is displaced or geo-masked to ensure household confidentiality.

The Bangladesh DHS has 19277 observations on children under 5 born between 2006 and 2018, with 6171 (32.01%) in 2011, 5875 (30.48%) in 2014 and 7231 (37.51%) in 2017-18 residing in 1871 clusters. The working sample is composed of children under age 5. I drop

children whose HAZ, WHZ and WAZ z scores are below -6 and above 6. This is in line with WHO's recommendation for the calculation of growth curve for healthy children by excluding values beyond biological interpretation (Ai and Tan 2025).

To capture child development status, the WHO Child Growth Standards calculates internationally recognized indicators including Height for Age (HAZ), Weight for Height (WHZ) and Weight for Age (WAZ) z scores for healthy children. The height for age z score used to determine stunting is calculated as follows:

Height for Age (HAZ) Z Score is defined as:

$$\text{HAZ} = \frac{\text{Actual Height of Child} - \text{Median Height of Reference Child Population}}{\text{Standard Deviation of Height of Reference Child Population}} \quad (1)$$

This score helps to capture the severity of chronic malnutrition. I also construct the variable of stunting whereby a child is considered to be stunted if $\text{HAZ} < -2 \text{ SD}$. For severe stunting, the cutoff is increased to $\text{HAZ} < -3 \text{ SD}$.

The weight-for-height z score for wasting is determined as follows:

$$\text{WHZ} = \frac{\text{Actual Weight of Child} - \text{Median Weight of Reference Child Population}}{\text{Standard Deviation of Weight of Reference Child Population}} \quad (2)$$

This measure reflects acute malnutrition due to short-term nutritional deficits.

The weight-for-age z score is calculated as follows:

$$\text{WAZ} = \frac{\text{Actual Weight of Child} - \text{Median Weight of Reference Child Population}}{\text{Standard Deviation of Weight of Reference Child Population}} \quad (3)$$

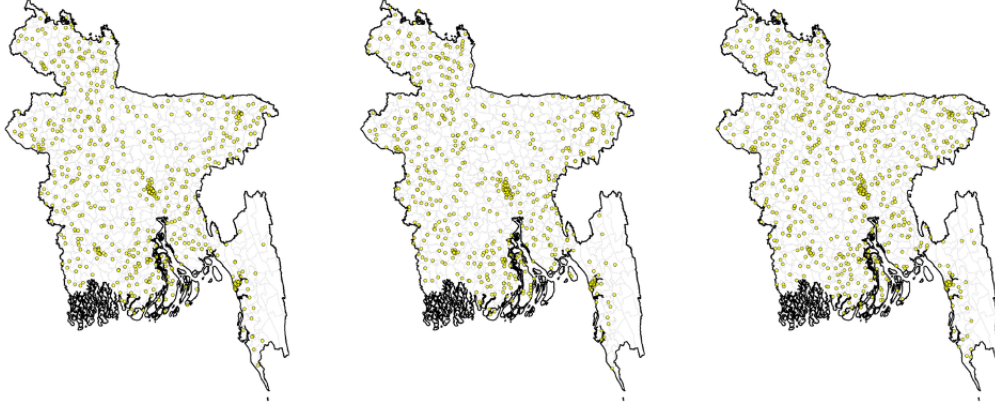


Figure 1: Sample of BDHS Households 2011–18

Note: The figure above plots clusters having children under 5 years of age across all survey waves.

BDHS also provides detailed information on various socioeconomic characteristics. I control for a standard set of household and individual characteristics in my analysis including child’s age, gender, mother’s education, whether the child resides in an urban area, household size and birth order of the child. Heterogeneity analysis is conducted using these characteristics as well as household wealth index to examine how child health differs across different socioeconomic groups.

3.2 Pollution Data

I utilize satellite pollution data obtained from Atmospheric Composition Analysis Group at Washington University in St. Louis (version V5.GL.05.02) (Van Donkelaar et al. 2021) to determine average pollution levels at the cluster level. Having a spatial resolution of $0.01^{\circ} \times 0.01^{\circ}$ (1 km at the equator), this data provides monthly pollution averages for each cluster in the sample. I use this data to calculate average lifetime pollution exposure for each child. This helps to capture the accumulated insult of pollution faced by each child over the course of his life and is in line with prior literature which also rely on lifetime pollution exposures (Balietti, Datta and Veljanoska 2022).

In order to determine the impact of lifetime pollution exposure on child health outcomes, I assume that the household does not migrate but instead resides in the same grid cell to

date in which the child was born. To address this concern, I examine the number of years at current residence reported by households in the survey. This information is only available in the 2017-18 round. 4% of mothers reported moving after the child’s birth, with 88% reporting they were interviewed at their usual residence.

Figure 2 illustrates the spatial distribution of average pollution levels from 2005 to 2018. Across all three periods 2005-11, 2008-14 and 2012-18, pollution levels are persistently higher in the southwestern regions of the country. Pollution sources include brick kilns, industrial activity and transboundary pollution transport. Over time, it can be seen that pollution levels increase nationwide indicating a substantial deterioration in air quality. This variation in exposure allows me to examine the effects of lifetime air pollution on child growth outcomes.

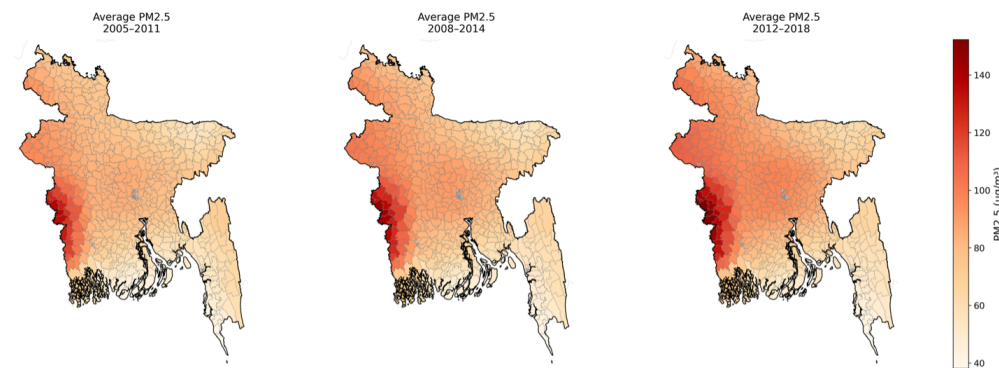


Figure 2: Average PM2.5 Across Survey Waves 2011–18

3.3 Thermal Inversions Data

Data for thermal inversions is taken from Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA2, version 5.12.4 M2I3NPASM) released by NASA. This dataset has a spatial resolution of $0.5^\circ \times 0.625^\circ$ (approximately 50x60 km grid) and is collected every 3 hours. I construct the measure of thermal inversions by counting the number of thermal inversions when the temperature in the first layer (1000hPa-110m) is lower than that in the second layer (975hPa-320m). If the difference is negative, I truncate it to 0. I aggregate the number of thermal inversions for lifetime exposure of the child at

cluster level. This is computed by calculating the difference between date of conception and date of interview. As a robustness check, I use thermal inversion strength which measures the magnitude of thermal inversions for each cluster. I construct this by calculating the temperature difference between second and first layer.

3.4 Weather Data

I utilize meteorological data from European Center for Medium-Range Weather Forecasts (ECMWF) ERA5 reanalysis data. This provides information on hourly basis at a high spatial resolution of $0.25^\circ \times 0.25^\circ$ grid (equivalent to 28×28 km at the equator). I aggregate lifetime average of children of these variables at the cluster level covering the period 2005-2018. These variables include wind speed, relative humidity (calculated using 2 m above ground temperature and dew point temperature), temperature and precipitation. Inclusion of these controls ensures atmospheric conditions experienced both during gestation and early childhood influencing fetal growth and postnatal development are accounted for (Misra and Kulshreshtha 2025).

Table 1 presents descriptive statistics for key variables used in the paper. The total number of children under 5 years of age across 3 waves of data is 19277, with 6171 (32.01%) in 2011, 5875 (30.48%) in 2014 and 7231 (37.51%) in 2017-18. Panel A summarizes child health indicators. Average HAZ score in the sample is -1.46 . This means that the average child in the sample is 0.54 ($2 - 1.46 = 0.54$) standard deviations away from the stunting threshold. Average WHZ score is -0.74 , so the average child is 1.26 ($2 - 0.74 = 1.26$) standard deviations away from the wasting threshold, with 12% of the children being wasted (Table 2). The average WAZ score is -1.35 , so the average child is 0.65 ($2 - 1.35 = 0.65$) standard deviations away from the underweight threshold, with 27.7% of the children in the sample being underweight.

Panel B summarizes household and demographic characteristics. The average child in the sample is 29 months of age, with nearly 23% of the children in the sample less than

a year old, 40% between 1 and 3 years old and 37% between 3 and 5 years of age. Male children form 51% of the sample. 41% of the children surveyed are first-born, with 32%, 16% and 11% being second, third and fourth born or of higher birth order. Average maternal education is 7 years. There are 6 household members on average, with a minimum of 2 and maximum of 31 members. 43% of the households have high wealth, with 20% being middle wealth and 37% being low wealth respectively.

Panel C reports descriptive statistics of pollution, thermal inversion and weather controls. Average lifetime pollution exposure is $61.6 \mu\text{g}/\text{m}^3$ and average number of thermal inversions experienced by children is 41 thermal inversions. Average temperature is 26 degrees, with high relative humidity and precipitation levels, warranting the inclusion of weather controls when estimating the impact of ambient air pollution on child growth.

Table 1: Descriptive Statistics

Variable	Obs.	Mean	Std. Dev.	Min	Max
Panel A: Child Health					
HAZ	19,277	-1.459	1.336	-5.94	5.88
Stunting	19,277	0.339	0.473	0	1
Severe Stunting	19,277	0.107	0.309	0	1
WHZ	19,277	-0.744	1.183	-5.00	4.80
Wasting	19,277	0.122	0.327	0	1
Severe Wasting	19,277	0.027	0.161	0	1
WAZ	19,277	-1.346	1.136	-5.83	4.74
Underweight	19,277	0.277	0.447	0	1
Panel B: Household and Demographic Characteristics					
Child Age (months)	19,277	29.033	17.328	0	59
Child Age (<1 year)	19,277	0.227	0.419	0	1
Child Age (1–3 years)	19,277	0.400	0.490	0	1
Child Age (3–5 years)	19,277	0.373	0.484	0	1
Male	19,277	0.514	0.500	0	1
Birth Order (1st)	19,277	0.408	0.492	0	1
Birth Order (2nd)	19,277	0.324	0.468	0	1
Birth Order (3rd)	19,277	0.159	0.366	0	1
Birth Order (4+)	19,277	0.109	0.311	0	1
Mother’s Education (years)	19,277	7.117	3.212	0	18
Household Size	19,277	6.097	2.830	2	31
Household Wealth (Low)	19,277	0.370	0.483	0	1
Household Wealth (Medium)	19,277	0.199	0.399	0	1
Household Wealth (High)	19,277	0.431	0.495	0	1
Urban	19,277	0.333	0.471	0	1
Panel C: Pollution, Thermal Inversions, and Weather					
PM2.5 ($\mu\text{g}/\text{m}^3$)	19,277	61.649	10.490	31.989	98.289
Thermal Inversions (2–1)	19,277	40.867	13.401	0	68.812
Wind Speed (m/s)	19,277	2.389	0.586	1.029	5.112
Relative Humidity (%)	19,277	79.795	2.052	71.512	86.283
Temperature ($^{\circ}\text{C}$)	19,277	25.645	0.754	19.532	28.149
Precipitation (mm)	19,277	200.609	69.929	80.952	681.669

Notes: Sample includes children born between 2006–2018 from BDHS 2011, 2014, and 2017–18. PM2.5 reports lifetime exposure from date of conception to date of interview, from the Washington University V5.GL.05.02 satellite product. Thermal inversions report the average number of inversions experienced. The number of observations corresponds to the total number of children under five years of age across survey waves.

Table 2 reports averages of child growth indicators by survey wave. Child health improves considerably over time across all outcomes. Stunting rates fall from 37.9% in 2011 to 30% by 2018 while severe stunting falls from 13.5% to 8.4% by 2018. Prevalence of wasting and severe wasting also decrease from 15% to 8.2% and 3.6% to 1.5% respectively over the same period. The likelihood of children being underweight also decreases considerably from 32.8%

in 2011 to 21% by 2018.

Table 2: Descriptive Statistics by Wave

Variable	BDHS 2011	BDHS 2014	BDHS 2017–18	Total
Stunting	37.87	34.06	30.31	33.87
Severe Stunting	13.50	10.45	8.39	10.66
Wasting	14.96	14.30	8.17	12.21
Severe Wasting	3.58	3.10	1.51	2.66
Underweight	32.81	30.43	21.05	27.68

3.5 Statistical Challenges

A major concern with determining the impact of cumulative pollution exposure on child growth outcomes is the endogeneity of ambient pollution levels. Because individuals sort into locations based on various attributes of the area including employment opportunities and local amenities, these may be correlated with environmental quality. Higher income, more educated and higher skilled households would prefer residential neighborhoods that have better air quality. However, these workers may also locate in areas which have higher economic activity and therefore higher pollution exposure. Because pollution may be correlated with unobserved household characteristics and local economic conditions that may affect child health, estimates may be biased (Suarez, Benatia and Thi 2025). Therefore, I exploit exogenous variation in air pollution generated by thermal inversions.

4 Empirical Analysis

4.1 Impact of Pollution on Child Health

I adopt an instrumental variable strategy to determine the causal impact of air pollution on child growth indicators as follows:

$$\begin{aligned} \text{PM}_{ict} = & \alpha_0 + \alpha_1 \text{TI}_{ict} + \boldsymbol{\alpha}_2^\top \mathbf{X}_{ict} + \boldsymbol{\alpha}_3^\top \mathbf{W}_{ict} \\ & + \theta_c + \delta_t + \varepsilon_{ict} \end{aligned} \quad (4)$$

$$\begin{aligned} Y_{ict} = & \beta_0 + \beta_1 \widehat{\text{PM}}_{ict} + \boldsymbol{\beta}_2^\top \mathbf{X}_{ict} + \boldsymbol{\beta}_3^\top \mathbf{W}_{ict} \\ & + \phi_c + \mu_t + \vartheta_{ict} \end{aligned} \quad (5)$$

where PM_{ict} and TI_{ict} denote lifetime PM2.5 and thermal-inversion exposure for child i in cluster c and birth cohort t . These are computed from three hourly (thermal inversions) and monthly (pollution) raw data aggregated at the grid cell monthly level and then averaged over the child's exposure window, which ranges from the month of conception to the month of interview. $\widehat{\text{PM}}_{ict}$ is the mean monthly PM2.5 concentration ($\mu\text{g}/\text{m}^3$) over this window, and TI_{ict} is the mean monthly count of thermal inversions. $\mathbf{W}_{ict}$ are cluster-specific monthly means averaged over the same window for weather controls including wind speed, relative humidity, temperature, and precipitation. $\mathbf{X}_{ict}$ is a vector of child- and household-level characteristics, including child age, gender, mother's education, urban residence, household size, and birth order. Y_{ict} includes child health outcomes such as height-for-age, weight-for-age, and weight-for-height z -scores, as well as stunting, severe stunting, wasting, severe wasting, and underweight, for child i in cluster c and birth cohort t . The coefficient β_1 identifies the causal effect of lifetime PM2.5 exposure on child growth under the exclusion restriction that, conditional on weather controls and fixed effects, TI_{ict} affects child health only through PM2.5. However, there is no restriction placed on the channels through which pollution can impact child health. β_1 therefore captures both the direct physiological pathway together

with the income pathway documented in Chapter 1. Because of this, it is interpreted as a total, reduced form effect rather than a biological dose-response estimate. Identification rests on within-cluster and within-cohort differences in birth timing. This means that children in the same cluster and cohort born in different months experience different levels of pollution and thermal inversions over their exposure windows. β_1 is estimated from this variation alone. θ_c and ϕ_c are cluster fixed effects; δ_t and μ_t are birth-cohort fixed effects, where t indexes birth cohort, absorbing cohort-specific shock histories. Cluster fixed effects absorb differences across locations in pollution, thermal inversions and local economic development. Birth cohort fixed effects absorb nationwide shocks common to children born in the same period. Standard errors are clustered at the thermal inversion-grid level to account for correlation in the error term within grids.

Table 3 reports IV FE regression estimates in Columns 2-9 for each of child growth indicators. Column 1 reports the first stage relationship between pollution and thermal inversions. A one standard deviation increase in thermal inversions is associated with a 3 $\mu\text{g}/\text{m}^3$ increase in PM2.5 levels ($0.2203 \times 13.4 = 3$). A 1 $\mu\text{g}/\text{m}^3$ increase in PM2.5 concentrations reduces average HAZ score by 0.014 standard deviations, significant at 10% level. A 1 $\mu\text{g}/\text{m}^3$ increase in pollution increases the probability of stunting by 0.48 percentage points, significant at the 10% level. A 1 $\mu\text{g}/\text{m}^3$ increase in pollution reduces the probability of wasting by 0.28 percentage points, significant at the 10% level. Because WAZ is unchanged at -0.0006 (not significant) and HAZ is declining (-0.0144, significant), the WHZ mechanically increases and lowers wasting by construction. Columns 4, 7, 8 and 9 report estimates for severe stunting, severe wasting, WAZ and underweight. These are not statistically significant.

Results vary considerably across prior literature. A one standard deviation increase in pollution levels increases the prevalence of stunting by 9-11% (Xu et al. 2024, Balietti et al. 2022), underweight by 12% and wasting by 8.6% (Balietti et al. 2022). Expressed as a share of the outcome standard deviation, a one standard deviation increase in PM2.5 accounts for approximately 11% of the HAZ standard deviation and 11% of the stunting

standard deviation.² My results for wasting are supported by prior literature which finds low or no adverse impact of pollution on wasting. This can be explained by pollution impacting chronic, height based growth outcomes, while wasting, being a symptom of acute malnutrition, is more susceptible to current pollution levels. Furthermore, wasting can be mitigated by taking steps such as improving dietary intake and controlling for infectious disease-associated symptoms including diarrhea (Singh, Alagarajan and Ladusingh 2015).

Table 3: Impact of Pollution on Child Health

VARIABLES	(1) PM2.5	(2) HAZ	(3) Stunting	(4) Severe Stunting	(5) WHZ	(6) Wasting	(7) Severe Wasting	(8) WAZ	(9) Underweight
Thermal Inversions	0.2203*** (0.0438)								
Predicted PM2.5		-0.0144* (0.0080)	0.0048* (0.0027)	0.0018 (0.0015)	0.0094* (0.0051)	-0.0028* (0.0015)	-0.0010 (0.0007)	-0.0006 (0.0052)	0.0009 (0.0022)
Observations	19,277	19,277	19,277	19,277	19,277	19,277	19,277	19,277	19,277
KP F stat	25.3								
Household controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors clustered at the thermal-inversion grid level

*** p<0.01, ** p<0.05, * p<0.1

4.2 Heterogeneity

In this section, I explore if my results are heterogeneous across various child, household and geographic characteristics. For this, I estimate the set of equations below:

$$PM_{ict} = \alpha_0 + \alpha_1 TI_{ict} + \alpha_2 (TI_{ict} \times Z_{ict}) + \alpha_3 Z_{ict} + \alpha_4^\top \mathbf{X}_{ict} + \alpha_5^\top \mathbf{W}_{ct} + \theta_c + \rho_t + \varepsilon_{ict} \quad (6)$$

$$Y_{ict} = \beta_0 + \beta_1 \widehat{PM}_{ict} + \beta_2 (\widehat{PM}_{ict} \times Z_{ict}) + \beta_3 Z_{ict} + \beta_4^\top \mathbf{X}_{ict} + \beta_5^\top \mathbf{W}_{ct} + \phi_c + \tau_t + \vartheta_{ict} \quad (7)$$

²My per- $\mu\text{g}/\text{m}^3$ coefficients are somewhat larger than those in Baliatti et al. (2022) (-0.0144 vs. -0.01104 for HAZ; 0.0048 vs. 0.00288 for stunting), but my sample has a lower PM2.5 standard deviation (10.5 vs. $17.0 \mu\text{g}/\text{m}^3$) and a lower HAZ standard deviation (1.336 vs. 1.587). As a result, the implied per-standard-deviation effects are very close to theirs: a one standard deviation increase in PM2.5 accounts for about 11% of the HAZ standard deviation (vs. 11.8% in their study) and about 11% of the stunting standard deviation (vs. 10.9%).

where Z_{ihct} denotes a vector of indicator variables for gender, age, birth order, household wealth and region of residence. β_1 captures the effect of pollution on child growth indicators for the omitted reference group. β_2 captures the direct effect of child and household characteristics on child health outcomes relative to the reference category. β_3 captures the differential impact of pollution on child health outcomes for the non-reference subgroup relative to the reference group. θ_c represent cluster fixed effects, ρ_t denote birth cohort fixed effects, and ϑ represents the error term. Standard errors are clustered at the thermal inversion grid level.

Gender Heterogeneity: A $1 \mu\text{g}/\text{m}^3$ increase in PM2.5 reduces the HAZ score of girls (the base group) by 0.013 standard deviations (significant at the 10% level). The $\text{PM2.5} \times \text{Male}$ interaction is small and statistically insignificant, indicating that boys are not differentially affected. Across all eight outcomes, the interaction of pollution with gender is statistically indistinguishable from zero, signifying that pollution does not differentially affect male and female children.

Age Heterogeneity: I then conduct heterogeneity analysis by age. The estimates determine how the effect of average lifetime pollution exposure differs across children observed at different ages and are not age specific dose-response effects. For the youngest children (0–12 months, the base group), a $1 \mu\text{g}/\text{m}^3$ increase in PM2.5 reduces the HAZ score by 0.013 standard deviations, raises the WHZ score by 0.016 standard deviations, and reduces the probability of wasting by 0.34 percentage points, all significant. The significant age interactions indicate that older children are differentially affected relative to the under-one base: for HAZ, the positive interactions for ages 13–36 months (0.0096*) and 37–60 months (0.0127**) imply that pollution’s adverse height effect is concentrated in the youngest children and attenuates with age. On the other hand, for WHZ and wasting, the interaction estimates of -0.0165 and 0.0028 respectively reflect that older children are more adversely affected on these weight-based measures.

Birth Order Heterogeneity: For first-born children (the base group), a $1 \mu\text{g}/\text{m}^3$ increase in PM2.5 reduces the HAZ score by 0.015 standard deviations and raises the probability of stunting by 0.5 percentage points (base coefficient 0.005**), both significant at the 5% level. Among the weight-based outcomes, wasting (-0.0035^{**}) is significant at 5% level and WHZ (0.0087^*) is significant at the 10% level; the severe-stunting, severe wasting, WAZ, and underweight estimates are not statistically significant. The birth-order interactions are largely insignificant, with the exception of a small positive $\text{PM2.5} \times \text{Birth order 2}$ effect on severe stunting (0.0016^{**}) and a positive $\text{PM2.5} \times \text{Birth order 3}$ effect on wasting (0.0018^*); the magnitudes of these differences are small, indicating limited heterogeneity across birth order.

Regional Heterogeneity: Estimates reveal limited regional heterogeneity. For rural children, a $1 \mu\text{g}/\text{m}^3$ increase in PM2.5 reduces the HAZ score by 0.015 standard deviations (-0.0151^*) and reduces the probability of wasting by 0.29 percentage points (-0.0029^*); other rural estimates are not statistically significant. The $\text{PM2.5} \times \text{Urban}$ interaction is small and statistically insignificant for all eight outcomes, indicating that urban and rural children are not differentially affected by pollution in these data.

Table 4: Heterogeneity in Pollution Effects on Child Health

	HAZ	Stunt.	Sev. Stunt.	WHZ	Wast.	Sev. Wast.	WAZ	Underwt.
<i>Panel A: Gender (base = Female)</i>								
Predicted PM2.5	−0.0130* (0.0077)	0.0046* (0.0027)	0.0014 (0.0016)	0.0081 (0.0052)	−0.0030** (0.0015)	−0.0009 (0.0006)	−0.0007 (0.0049)	0.0011 (0.0022)
PM2.5 × Male	−0.0027 (0.0029)	0.0004 (0.0012)	0.0007 (0.0007)	0.0023 (0.0022)	0.0005 (0.0006)	−0.0001 (0.0003)	0.0002 (0.0025)	−0.0004 (0.0008)
<i>N</i>	19277	19277	19277	19277	19277	19277	19277	19277
<i>Panel B: Age (base = 0–12 mo)</i>								
Predicted PM2.5	−0.0131** (0.0066)	0.0043* (0.0023)	0.0023* (0.0012)	0.0161*** (0.0048)	−0.0034** (0.0014)	−0.0008 (0.0006)	0.0029 (0.0051)	−0.0003 (0.0018)
PM2.5 × Age 13–36 mo	0.0096* (0.0056)	−0.0024 (0.0017)	−0.0021* (0.0012)	−0.0008 (0.0034)	−0.0008 (0.0009)	−0.0007* (0.0004)	0.0047 (0.0039)	−0.0003 (0.0013)
PM2.5 × Age 37–60 mo	0.0127** (0.0056)	−0.0031* (0.0017)	−0.0023** (0.0011)	−0.0165*** (0.0056)	0.0028*** (0.0010)	0.0001 (0.0005)	−0.0038 (0.0041)	0.0012 (0.0012)
<i>N</i>	19277	19277	19277	19277	19277	19277	19277	19277
<i>Panel C: Birth Order (base = first born)</i>								
Predicted PM2.5	−0.0147** (0.0074)	0.0050** (0.0023)	0.0016 (0.0014)	0.0087* (0.0051)	−0.0035** (0.0015)	−0.0010 (0.0007)	−0.0006 (0.0050)	0.0017 (0.0021)
PM2.5 × Birth order 2	0.0002 (0.0027)	0.0007 (0.0011)	0.0016** (0.0007)	0.0030 (0.0032)	0.0008 (0.0009)	−0.0001 (0.0004)	0.0015 (0.0024)	−0.0012 (0.0011)
PM2.5 × Birth order 3	0.0019 (0.0038)	−0.0020 (0.0022)	−0.0001 (0.0011)	−0.0017 (0.0037)	0.0018* (0.0010)	0.0007 (0.0006)	−0.0012 (0.0029)	−0.0017 (0.0015)
PM2.5 × Birth order 4+	−0.0011 (0.0039)	0.0001 (0.0013)	−0.0012 (0.0015)	0.0010 (0.0035)	0.0012 (0.0009)	−0.0008 (0.0006)	−0.0005 (0.0036)	−0.0007 (0.0026)
<i>N</i>	19277	19277	19277	19277	19277	19277	19277	19277
<i>Panel D: Urban (base = Rural)</i>								
Predicted PM2.5	−0.0151* (0.0080)	0.0045 (0.0028)	0.0017 (0.0015)	0.0081 (0.0053)	−0.0029* (0.0015)	−0.0010 (0.0007)	−0.0015 (0.0052)	0.0015 (0.0022)
PM2.5 × Urban	0.0027 (0.0051)	0.0015 (0.0014)	0.0003 (0.0011)	0.0055 (0.0035)	0.0007 (0.0009)	−0.0001 (0.0005)	0.0042 (0.0039)	−0.0027 (0.0017)
Observations	19277	19277	19277	19277	19277	19277	19277	19277
R-squared	0.0664	0.0459	0.0239	0.0149	0.0044	0.0046	0.0712	0.0349

Robust standard errors clustered at the thermal-inversion grid level in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4.3 Welfare Implications

Next, I quantify the long-run economic costs of pollution exposure by determining how pollution-induced reduction in child height translates into losses in adult height and earnings. Following Baliatti, Datta and Veljanoska (2022), I determine this as follows:

$$\Delta H_a = \sigma_a \rho_a \hat{\beta}_1 \Delta P. \quad (9)$$

where H_a stands for adult height, σ_a denotes the standard deviation of adult height for 19-year-old men ($\sigma_a = 6.5$ cm) and women ($\sigma_a = 6.04$ cm) in the WHO reference population,

and ρ_a is the correlation between child and adult height for men and women taken from Molinari et al. (1995). $\hat{\beta}_1$ is the baseline IV FE estimate of the effect of a 1 $\mu\text{g}/\text{m}^3$ increase in lifetime PM2.5 exposure on the HAZ score (-0.0144 , Column 2 of Table ??) and ΔP is a one standard deviation increase in lifetime pollution exposure ($10.49 \mu\text{g}/\text{m}^3$). A one standard deviation increase in lifetime PM2.5 exposure therefore lowers the HAZ score by 0.151 standard deviations.

Next, I determine the impact on adult earnings using the equation below:

$$\Delta E_a = \left(\frac{\Delta H_a}{10} \right) \phi_g \times 100. \quad (10)$$

where ϕ_g is the height premium per 10 cm from Sohn (2015) (7.5% for males and 13% for females). This is determined using Indonesian data, the most comparable country to Bangladesh. I then multiply this by 100 to convert it into a percentage.

Table 5 reports the estimates pertaining to losses in adult height and earnings due to early-life pollution exposure. Male children experience larger reductions in adult height than their female counterparts (0.58–0.80 cm versus 0.47–0.61 cm), reflecting both the higher standard deviation of adult male height and the stronger child–adult height correlation for boys. The implied losses rise with the age at which the child is observed, since ρ_a strengthens with age; for girls the profile flattens after age three, as the child–adult height correlation for females plateaus at approximately 0.67. On the other hand, females face larger earning penalties as compared to males, with a one standard deviation increase in pollution lowering lifetime earnings by 0.60–0.79% for women as against 0.43–0.60% for men. This is due to the higher height premium for women of 13% per 10 cm as compared to 7.5% per 10 cm for men (Sohn 2015).

These calculations apply the pooled baseline estimate at every age. The age heterogeneity reported in Table 4 indicates that the adverse HAZ effect is concentrated among children under one year and attenuates thereafter, so the age profile below should be read as reflecting

the strengthening child–adult height correlation rather than an age-varying dose-response relationship.

Table 5: Impact of Pollution Exposure on Adult Height and Earnings

Age (years)	ρ_a Males	ρ_a Females	ΔH_a Males (cm)	ΔH_a Females (cm)	ΔE_a Males (%)	ΔE_a Females (%)
0	0.59	0.51	-0.58	-0.47	-0.43	-0.60
1	0.70	0.59	-0.69	-0.54	-0.52	-0.70
2	0.74	0.65	-0.73	-0.59	-0.54	-0.77
3	0.76	0.67	-0.75	-0.61	-0.56	-0.79
4	0.79	0.67	-0.78	-0.61	-0.58	-0.79
5+	0.81	0.67	-0.80	-0.61	-0.60	-0.79

Notes: Estimates are based on Equations (9) and (10). $\hat{\beta}_1 = -0.0144$ is the baseline IV FE effect of a $1 \mu\text{g}/\text{m}^3$ increase in lifetime PM2.5 exposure on the HAZ score and $\Delta P = 10.49 \mu\text{g}/\text{m}^3$ is one standard deviation of lifetime PM2.5 exposure in the sample. σ_a equals 6.5 cm for men and 6.04 cm for women in the WHO reference population of 19-year-olds. ρ_a is the correlation between a child’s current height and their adult height from Molinari et al. (1995), as reported in Cole and Wright (2011), averaged over each completed year of age. ϕ_g is the height premium per 10 cm from Sohn (2015) (7.5% for males, 13% for females).

Figure 3 plots out the loss in adult height (Panel A) and in adult earnings (Panel B) by gender and age. Panel A shows males experience a greater reduction in height at every age as compared to females. In Panel B, this trend is reversed where females experience greater earning losses as compared to their male counterparts at every age.

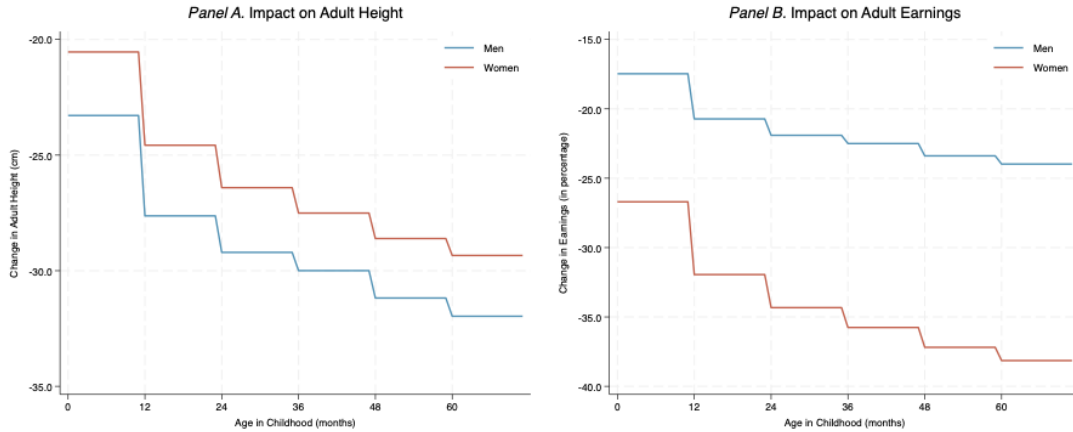


Figure 3: Losses in Adult Height and Earnings Due to Early-Life Pollution Exposure

4.4 Robustness

I next validate my results by conducting a series of robustness checks. The first row reports the baseline estimate, while the subsequent rows report estimates from these checks. I first test for sensitivity to treatment of meteorological controls by either controlling for them flexibly or excluding them altogether. Including weather controls in deciles leaves the height-based results intact: a $1 \mu\text{g}/\text{m}^3$ increase in PM2.5 reduces the HAZ score by 0.015 standard deviations and raises the probability of stunting by 0.58 percentage points. When weather controls are excluded, HAZ and stunting estimates lose significance. While weather controls are correlated with thermal inversions, they also impact child health directly through the channels of disease environment, nutrition and thermal stress. This means the exclusion restriction only holds conditional on weather controls. Excluding them from the specification is uninformative about the impact of pollution on child growth.

I then winsorize my child health outcomes to ensure the results are not driven by outliers. The estimates remain close to the baseline: a $1 \mu\text{g}/\text{m}^3$ increase in PM2.5 reduces the HAZ score by 0.015 standard deviations and raises the probability of stunting by 0.48 percentage points, with wasting also significant as in the baseline. I also include birth-month fixed effects to account for the effect of season of birth on child health and to remove any concern of it confounding thermal inversions. I find that my estimates remain unchanged (HAZ -0.0165^{**}). I then address the concern that lifetime pollution exposure is mechanically equal to the child's age and nine months of gestation. Therefore, I reconstruct both pollution and thermal inversions over fixed length windows tied to the timing of pregnancy. The gestational window is 9 months long for each child in the sample. Under this window, I find that a $1 \mu\text{g}/\text{m}^3$ increase in gestational PM2.5 reduces HAZ score by 0.004 standard deviations (-0.0036^{***}) and raises the probability of stunting by 0.09 percentage points (0.0009^{**}), both significant, while the weight-based outcomes remain largely statistically insignificant. This pattern is consistent with the fetal origins hypothesis, under which pregnancy and the first 1,000 days are critical windows for growth (Almond and Currie 2011; Ai and Tan 2025).

I then restrict my sample to children who are at least 24 months old at interview since they fall into the first 1000 days window (from conception to 24 months). I find that the estimate is negative but statistically insignificant (-0.0092 for HAZ). In the last row, I re-estimate the baseline specification on the same restricted sample to see if the attenuation in results reflects the change in window or the change in sample. I find that this estimate is positive and insignificant and is indicative of the attenuation being driven by the restricted sample instead of window exposure. Taken together, these checks provide evidence that the baseline estimates are robust and not sensitive to alternative specifications.

Table 6: Robustness Checks

	(1) HAZ	(2) Stunting	(3) Severe Stunting	(4) WHZ	(5) Wasting	(6) Severe Wasting	(7) WAZ	(8) Underweight
Baseline	-0.0144* (0.0080)	0.0048* (0.0027)	0.0018 (0.0015)	0.0094* (0.0051)	-0.0028* (0.0015)	-0.0010 (0.0007)	-0.0006 (0.0052)	0.0009 (0.0022)
Controlling flexibly for meteorological controls	-0.0154* (0.0084)	0.0058** (0.0029)	0.0037** (0.0019)	0.0147*** (0.0049)	-0.0032** (0.0013)	-0.0013* (0.0007)	0.0027 (0.0054)	0.0017 (0.0025)
Excluding Meteorological Controls	-0.0029 (0.0039)	0.0006 (0.0013)	-0.0002 (0.0009)	0.0054** (0.0027)	-0.0015** (0.0007)	-0.0006* (0.0003)	0.0037 (0.0031)	-0.0009 (0.0011)
Winsorized Child Health Indicators	-0.0151* (0.0079)	0.0048* (0.0027)	0.0018 (0.0015)	0.0087* (0.0048)	-0.0028* (0.0015)	-0.0010 (0.0007)	-0.0008 (0.0051)	0.0009 (0.0022)
Including Birth-Month FE (season of birth)	-0.0165* (0.0094)	0.0053* (0.0031)	0.0020 (0.0017)	0.0103* (0.0055)	-0.0029* (0.0017)	-0.0010 (0.0007)	-0.0009 (0.0058)	0.0013 (0.0024)
Gestational window (conception to birth, 9 months)	-0.0036*** (0.0011)	0.0009** (0.0004)	0.0006** (0.0003)	-0.0007 (0.0010)	-0.0001 (0.0003)	-0.0001 (0.0001)	-0.0020** (0.0009)	0.0006 (0.0004)
First 1,000 days (conception to 24 months, 33 months)	-0.0092 (0.0075)	0.0044* (0.0025)	0.0021 (0.0019)	-0.0031 (0.0059)	-0.0019 (0.0016)	-0.0010 (0.0007)	-0.0083 (0.0059)	0.0019 (0.0026)
Cumulative exposure, first-1,000-days sample	0.0043 (0.0121)	0.0026 (0.0045)	0.0017 (0.0032)	-0.0040 (0.0097)	-0.0028 (0.0030)	-0.0009 (0.0012)	-0.0013 (0.0099)	-0.0009 (0.0041)
Observations	19,277	19,277	19,277	19,277	19,277	19,277	19,277	19,277

Each row reports the coefficient on instrumented PM2.5 from a separate specification. Rows 1–5 use cumulative exposure from conception to interview; rows 6–7 use fixed-length exposure windows tied to the timing of pregnancy. The gestational window is available for every child; the first-1,000-days window requires children to be at least 24 months old at interview (row 8 re-estimates the cumulative measure on that restricted sample). Standard errors clustered at the thermal-inversion grid level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

5 Conclusion

This paper examines the impact of lifetime exposure to fine particulate matter on child growth in Bangladesh, a country characterized by high pollution levels and poor child health outcomes. To my knowledge, it provides the first causal estimates for Bangladesh, instrumenting pollution with thermal inversions and measuring exposure cumulatively from conception to interview across a full set of child anthropometric indicators. Exposure during gestation alone has a statistically significant effect on height-based outcomes, consistent with the fetal origins hypothesis, while the weaker estimate over the first 1,000 days appears to be driven by the restricted sample that window requires rather than by the exposure window itself.

Findings reveal that a $1 \mu\text{g}/\text{m}^3$ increase in pollution levels reduces the height-for-age (HAZ) score by 0.014 standard deviations and increases the probability of stunting by 0.48 percentage points. The estimates for severe stunting, weight-for-age, severe wasting and underweight are not statistically significant. Weight-for-height (WHZ) rises and wasting falls, but this pattern is mechanical: because weight-for-age is unchanged while height-for-age declines, weight measured relative to height increases by construction. Taken together, the results indicate that cumulative pollution exposure impairs chronic, height-based growth rather than acute, weight-based nutritional status. These findings are robust across a range of alternative specifications, including flexible meteorological controls, winsorized outcomes, birth-month fixed effects, and fixed-length exposure windows tied to the timing of pregnancy.

Consistent with prior literature, I find pollution exposure impairing linear, height-based growth (Balietti, Datta, and Veljanoska 2022; Xu et al. 2024; Goyal and Canning 2018; Sinharoy et al. 2020). In contrast to Balietti, Datta and Veljanoska (2022), who report that pollution raises wasting, I find no evidence of an adverse effect on weight-based outcomes, consistent with the view that weight-based deficits are determined more by current nutrition and infection than by cumulative pollution exposure (Singh, Alagarajan and Ladusingh 2015). In contrast to Kurata et al. (2020), who report that prenatal ambient PM_{2.5} ex-

posure is associated with stunting in boys but not girls in Bangladesh, I do not find any heterogeneous pollution response by gender.

Applying the framework of Baliatti, Datta and Veljanoska (2022), I translate these growth deficits into long-run economic costs. A one standard deviation increase in lifetime pollution exposure ($10.5 \mu\text{g}/\text{m}^3$) implies a reduction in adult height of 0.58–0.80 cm for men and 0.47–0.61 cm for women, and a corresponding fall in lifetime earnings of 0.43–0.60% and 0.60–0.79% respectively. Women face larger earnings losses despite smaller height deficits, since the height premium is substantially higher for women than for men (Sohn 2015). These are back-of-the-envelope projections rather than directly estimated long-run effects, but they indicate that the growth deficits documented here carry measurable economic consequences in adulthood.

This paper also faces several limitations. A key limitation of this study is the lack of longitudinal data which would allow me to investigate the impact of cumulative lifetime pollution exposure for the same children as they age. Another limitation is the lack of information on time spent outdoors by mothers and children and any pollution avoidance or defensive behaviors that may be adopted by households. Relatedly, my welfare calculations rely on a height premium estimated for Indonesia and on childhood-to-adult height correlations drawn from a reference population, so they should be read as illustrative of magnitude rather than as precise estimates for Bangladesh.

Findings from this paper provide valuable empirical insights into the importance of environmental conditions in shaping early-life health outcomes and highlight the need for enforcing sustained environmental policies that target pollution effectively so that child health outcomes can be optimized. Because early-life growth deficits carry measurable costs in adulthood, sustained reductions in ambient pollution offer returns not only in child health but in future labor market productivity. Concerted efforts to lower pollution levels therefore represent an investment in human capital whose returns accrue over the working lives of the children affected.

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Appendix A

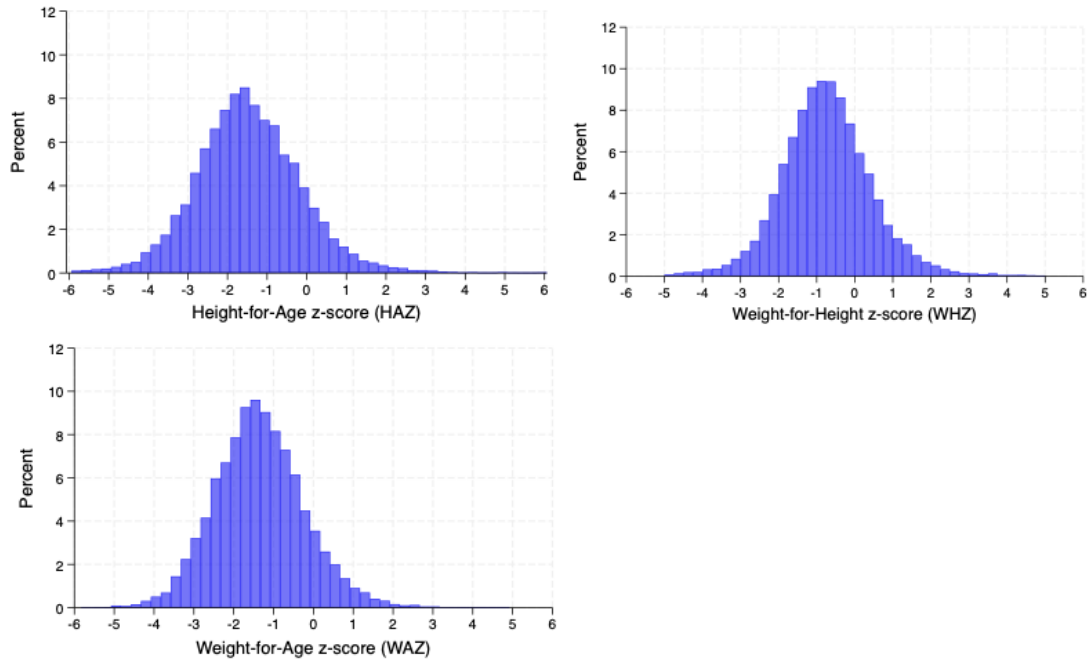


Figure A1: Distribution of Child Growth Indicators

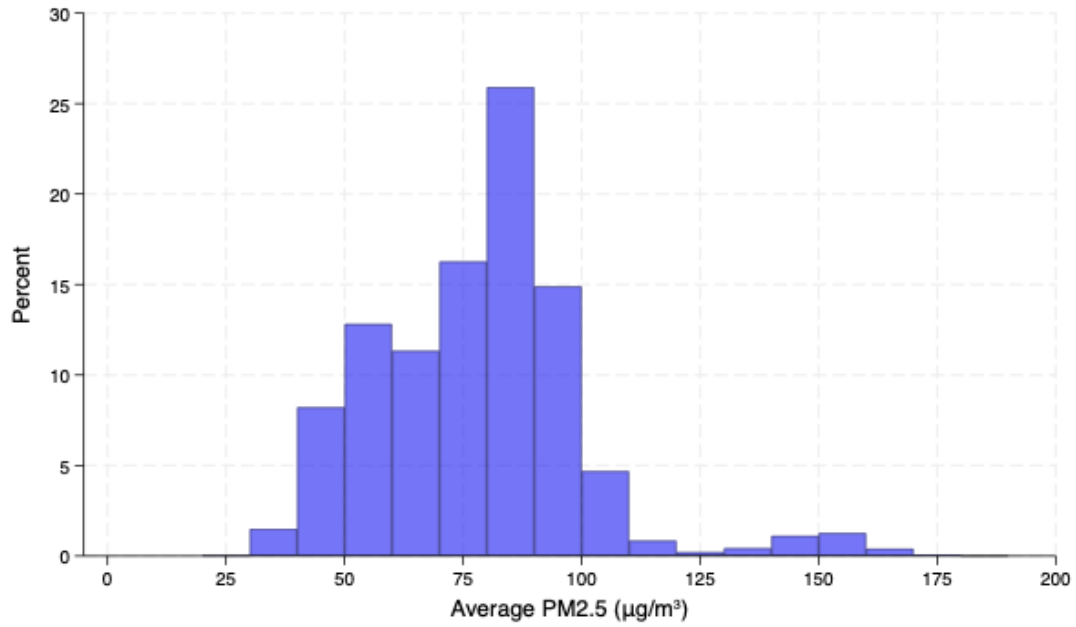


Figure A2: Distribution of Average Lifetime PM2.5

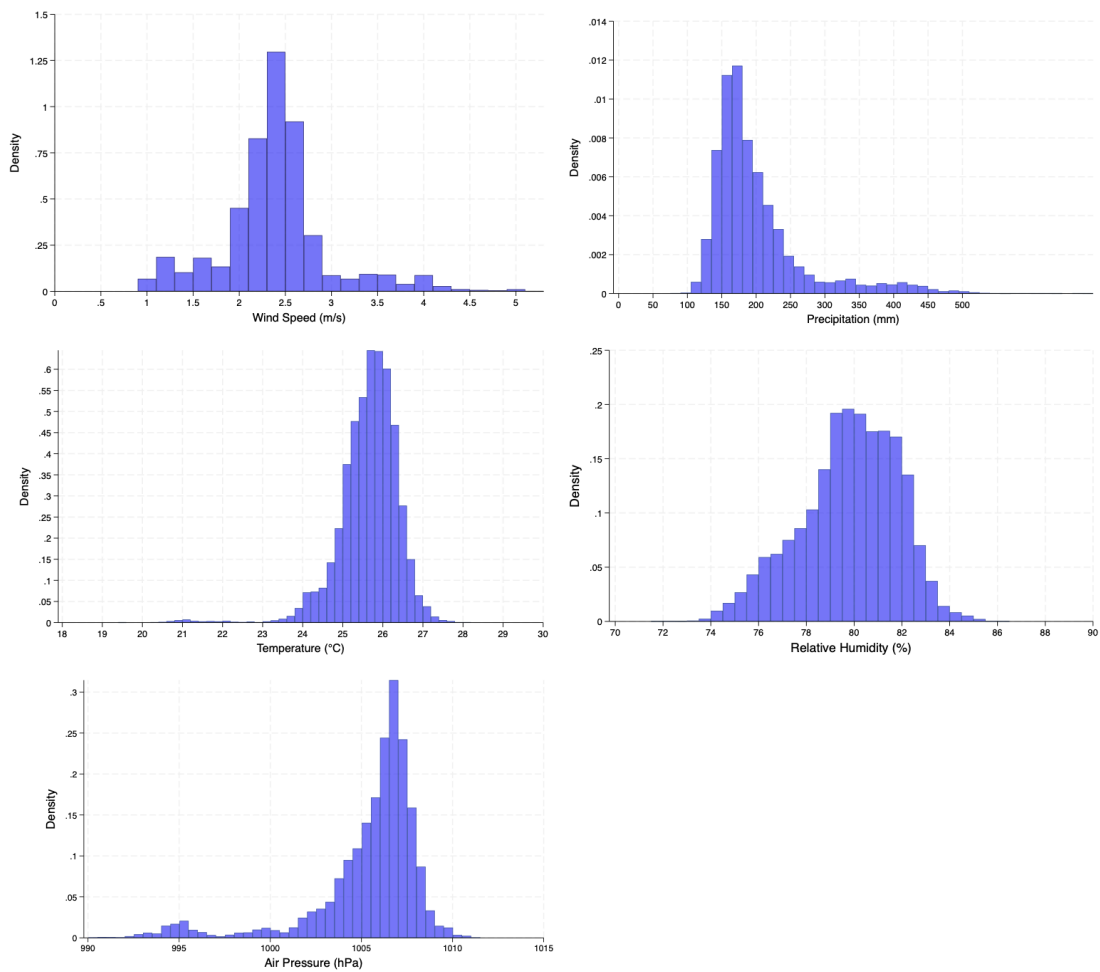
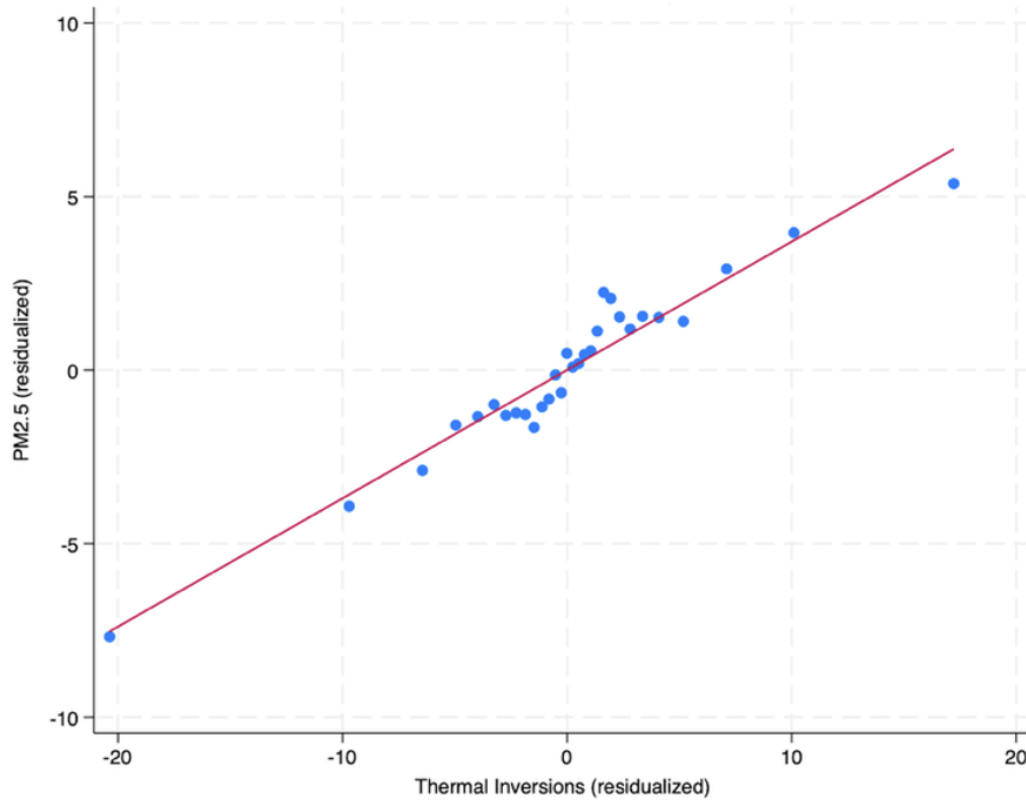


Figure A3: Distribution of Weather Variables



Note: Binned scatter plots with 30 quantile bins. Variables residualized on controls, cluster and wave fixed effects.

Figure A4: First Stage Relationship Between Thermal Inversions and PM2.5

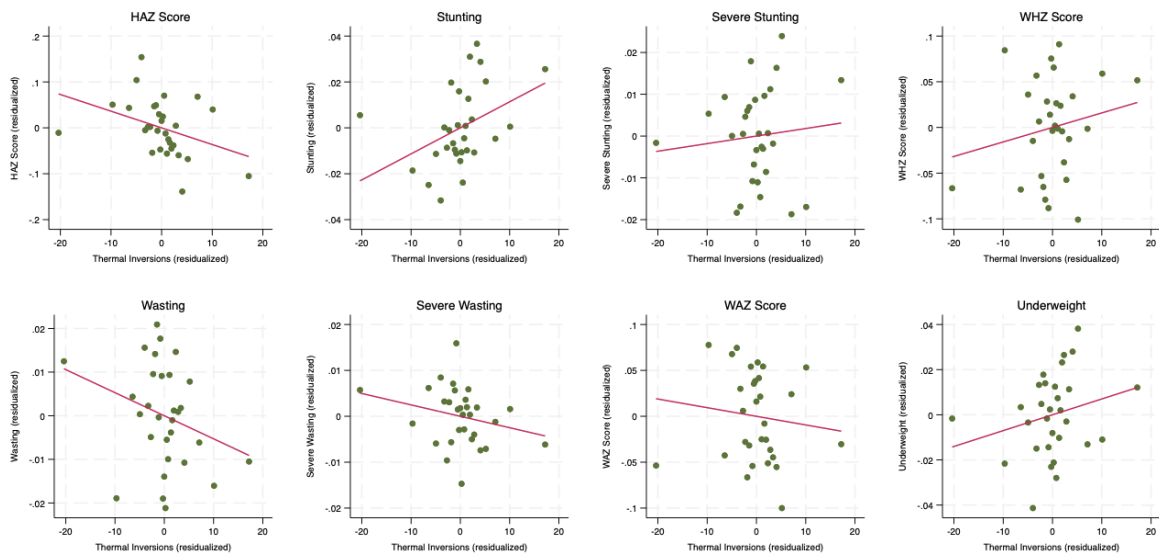


Figure A5: Reduced Form: Thermal Inversions and Child Health Outcomes

Appendix B

Table B1: Impact of Pollution on Child Health, Full Coefficients

	HAZ	Stunt.	Sev. Stunt.	WHZ	Wast.	Sev. Wast.	WAZ	Underwt.
Predicted PM2.5	-0.0144* (0.0080)	0.0048* (0.0027)	0.0018 (0.0015)	0.0094* (0.0051)	-0.0028* (0.0015)	-0.0010 (0.0007)	-0.0006 (0.0052)	0.0009 (0.0022)
Male	0.0050 (0.0175)	0.0064 (0.0078)	-0.0008 (0.0044)	-0.0181 (0.0150)	0.0129*** (0.0047)	0.0061** (0.0025)	0.0354** (0.0164)	-0.0103 (0.0066)
Child age (months)	-0.0076*** (0.0010)	0.0012*** (0.0004)	-0.0003* (0.0002)	-0.0038*** (0.0008)	-0.0010*** (0.0002)	-0.0007*** (0.0001)	-0.0070*** (0.0007)	0.0015*** (0.0003)
Mother's education (years)	0.0640*** (0.0044)	-0.0171*** (0.0013)	-0.0080*** (0.0008)	0.0217*** (0.0029)	-0.0023*** (0.0008)	-0.0006 (0.0005)	0.0529*** (0.0036)	-0.0128*** (0.0011)
Household size	-0.0035 (0.0044)	0.0021 (0.0014)	0.0007 (0.0010)	0.0034 (0.0039)	-0.0004 (0.0011)	-0.0002 (0.0005)	-0.0007 (0.0044)	0.0004 (0.0014)
Birth order	0.0111 (0.0075)	0.0021 (0.0028)	0.0032* (0.0018)	-0.0129** (0.0062)	-0.0004 (0.0024)	0.0011 (0.0013)	-0.0014 (0.0059)	0.0000 (0.0027)
Wealth: medium	0.1306*** (0.0366)	-0.0625*** (0.0106)	-0.0341*** (0.0074)	0.0387 (0.0237)	-0.0060 (0.0066)	-0.0025 (0.0034)	0.1099*** (0.0256)	-0.0488*** (0.0102)
Wealth: high	0.4194*** (0.0349)	-0.1538*** (0.0140)	-0.0691*** (0.0080)	0.1716*** (0.0240)	-0.0227*** (0.0066)	-0.0040 (0.0032)	0.3645*** (0.0229)	-0.1234*** (0.0089)
Urban	0.0988*** (0.0374)	-0.0064 (0.0102)	-0.0008 (0.0068)	0.0184 (0.0289)	0.0057 (0.0079)	0.0069* (0.0036)	0.0649** (0.0297)	-0.0049 (0.0121)
Wind speed (m/s)	-0.0375 (0.0529)	0.0148 (0.0139)	0.0248** (0.0118)	-0.1745*** (0.0306)	0.0310*** (0.0078)	0.0120*** (0.0045)	-0.1415*** (0.0393)	0.0425*** (0.0139)
Relative humidity (%)	-0.0277* (0.0153)	0.0116** (0.0051)	0.0120*** (0.0024)	0.0285** (0.0144)	-0.0038 (0.0039)	-0.0020 (0.0018)	0.0031 (0.0123)	0.0021 (0.0048)
Temperature (C)	0.0403 (0.0329)	-0.0212* (0.0112)	-0.0116 (0.0073)	0.0747*** (0.0246)	-0.0223*** (0.0066)	-0.0049 (0.0035)	0.0736*** (0.0266)	-0.0316*** (0.0090)
Precipitation (mm)	-0.0012* (0.0007)	0.0004 (0.0003)	0.0001 (0.0001)	-0.0002 (0.0006)	-0.0000 (0.0002)	-0.0000 (0.0001)	-0.0009* (0.0005)	0.0003 (0.0002)
Observations	19,277	19,277	19,277	19,277	19,277	19,277	19,277	19,277
Cluster FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

As Table 3, with all control coefficients reported. Standard errors clustered at the thermal-inversion grid level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B2: First Stage, Full Coefficients

	PM2.5
Thermal Inversions	0.2203*** (0.0438)
Male	-0.1161* (0.0594)
Child age (months)	0.0160* (0.0094)
Mother's education (years)	-0.0797*** (0.0177)
Household size	0.0422* (0.0227)
Birth order	-0.0371 (0.0289)
Wealth: medium	0.6480*** (0.1475)
Wealth: high	1.3938*** (0.2832)
Urban	1.8309*** (0.5276)
Wind speed (m/s)	4.3486*** (1.0392)
Relative humidity (%)	0.4406 (0.3087)
Temperature (C)	-2.4140*** (0.5579)
Precipitation (mm)	-0.0516*** (0.0085)
Observations	19,277
Cluster FE	Yes
Cohort FE	Yes

First-stage regression of lifetime PM2.5 exposure on lifetime thermal-inversion exposure, with all control coefficients reported. Standard errors clustered at the thermal-inversion grid level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B3: Heterogeneity: Gender, Full Coefficients

	HAZ	Stunt.	Sev. Stunt.	WHZ	Wast.	Sev. Wast.	WAZ	Underwt.
Predicted PM2.5	-0.0130*	0.0046*	0.0014	0.0081	-0.0030**	-0.0009	-0.0007	0.0011
	(0.0077)	(0.0027)	(0.0016)	(0.0052)	(0.0015)	(0.0006)	(0.0049)	(0.0022)
PM2.5 × Male	-0.0027	0.0004	0.0007	0.0023	0.0005	-0.0001	0.0002	-0.0004
	(0.0029)	(0.0012)	(0.0007)	(0.0022)	(0.0006)	(0.0003)	(0.0025)	(0.0008)
Male	0.1725	-0.0190	-0.0469	-0.1569	-0.0149	0.0138	0.0237	0.0139
	(0.1857)	(0.0760)	(0.0417)	(0.1349)	(0.0369)	(0.0183)	(0.1573)	(0.0519)
Child Age	-0.0076***	0.0012***	-0.0003*	-0.0038***	-0.0010***	-0.0007***	-0.0070***	0.0015***
	(0.0010)	(0.0004)	(0.0002)	(0.0008)	(0.0002)	(0.0001)	(0.0007)	(0.0003)
Mother's Educ.	0.0640***	-0.0171***	-0.0079***	0.0217***	-0.0023***	-0.0006	0.0529***	-0.0128***
	(0.0044)	(0.0013)	(0.0008)	(0.0029)	(0.0008)	(0.0005)	(0.0036)	(0.0011)
Household Size	-0.0034	0.0021	0.0007	0.0034	-0.0004	-0.0002	-0.0007	0.0004
	(0.0044)	(0.0014)	(0.0010)	(0.0039)	(0.0011)	(0.0005)	(0.0044)	(0.0014)
Birth Order	0.0111	0.0021	0.0032*	-0.0129**	-0.0004	0.0011	-0.0014	0.0000
	(0.0075)	(0.0028)	(0.0018)	(0.0062)	(0.0024)	(0.0013)	(0.0059)	(0.0027)
Wealth (Medium)	0.1305***	-0.0625***	-0.0341***	0.0387	-0.0060	-0.0025	0.1099***	-0.0488***
	(0.0367)	(0.0106)	(0.0074)	(0.0237)	(0.0066)	(0.0034)	(0.0256)	(0.0102)
Wealth (High)	0.4191***	-0.1537***	-0.0690***	0.1718***	-0.0227***	-0.0040	0.3645***	-0.1234***
	(0.0349)	(0.0140)	(0.0080)	(0.0241)	(0.0066)	(0.0032)	(0.0229)	(0.0089)
Urban	0.0984***	-0.0063	-0.0007	0.0187	0.0058	0.0068*	0.0649**	-0.0050
	(0.0373)	(0.0102)	(0.0068)	(0.0288)	(0.0078)	(0.0036)	(0.0296)	(0.0121)
Wind Speed	-0.0376	0.0148	0.0249**	-0.1744***	0.0310***	0.0120***	-0.1415***	0.0425***
	(0.0529)	(0.0138)	(0.0118)	(0.0305)	(0.0079)	(0.0045)	(0.0392)	(0.0139)
Rel. Humidity	-0.0277*	0.0116**	0.0120***	0.0285**	-0.0038	-0.0020	0.0031	0.0021
	(0.0153)	(0.0051)	(0.0024)	(0.0144)	(0.0039)	(0.0018)	(0.0123)	(0.0048)
Temperature	0.0404	-0.0212*	-0.0117	0.0746***	-0.0223***	-0.0049	0.0735***	-0.0316***
	(0.0331)	(0.0112)	(0.0073)	(0.0244)	(0.0066)	(0.0035)	(0.0265)	(0.0089)
Precipitation	-0.0012*	0.0004	0.0001	-0.0002	-0.0000	-0.0000	-0.0009*	0.0003
	(0.0007)	(0.0003)	(0.0001)	(0.0006)	(0.0002)	(0.0001)	(0.0005)	(0.0002)
Observations	19,277	19,277	19,277	19,277	19,277	19,277	19,277	19,277
Cluster FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Full-control version of the gender heterogeneity panel. Standard errors clustered at the thermal-inversion grid level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B4: Heterogeneity: Age, Full Coefficients

	HAZ	Stunt.	Sev. Stunt.	WHZ	Wast.	Sev. Wast.	WAZ	Underwt.
Predicted PM2.5	-0.0131** (0.0066)	0.0043* (0.0023)	0.0023* (0.0012)	0.0161*** (0.0048)	-0.0034** (0.0014)	-0.0008 (0.0006)	0.0029 (0.0051)	-0.0003 (0.0018)
PM2.5 × Age 13-36 mo	0.0096* (0.0056)	-0.0024 (0.0017)	-0.0021* (0.0012)	-0.0008 (0.0034)	-0.0008 (0.0009)	-0.0007* (0.0004)	0.0047 (0.0039)	-0.0003 (0.0013)
PM2.5 × Age 37-60 mo	0.0127** (0.0056)	-0.0031* (0.0017)	-0.0023** (0.0011)	-0.0165*** (0.0056)	0.0028*** (0.0010)	0.0001 (0.0005)	-0.0038 (0.0041)	0.0012 (0.0012)
Age 13-36 mo	-1.3296*** (0.3410)	0.3352*** (0.1046)	0.2027*** (0.0740)	-0.0369 (0.2110)	0.0245 (0.0546)	0.0303 (0.0273)	-0.6361*** (0.2314)	0.1125 (0.0787)
Age 37-60 mo	-1.3769*** (0.3369)	0.3223*** (0.1054)	0.1633** (0.0694)	0.8821** (0.3564)	-0.2138*** (0.0601)	-0.0339 (0.0326)	-0.1321 (0.2550)	0.0097 (0.0767)
Male	0.0033 (0.0174)	0.0068 (0.0078)	-0.0007 (0.0044)	-0.0189 (0.0152)	0.0129*** (0.0048)	0.0061** (0.0026)	0.0342** (0.0164)	-0.0100 (0.0066)
Mother's Educ.	0.0636*** (0.0044)	-0.0170*** (0.0013)	-0.0079*** (0.0008)	0.0221*** (0.0029)	-0.0024*** (0.0009)	-0.0006 (0.0005)	0.0528*** (0.0036)	-0.0128*** (0.0011)
Household Size	-0.0039 (0.0042)	0.0023* (0.0014)	0.0009 (0.0010)	0.0034 (0.0039)	-0.0002 (0.0011)	-0.0001 (0.0005)	-0.0007 (0.0044)	0.0004 (0.0014)
Birth Order	0.0120 (0.0076)	0.0018 (0.0028)	0.0030 (0.0019)	-0.0133** (0.0062)	-0.0004 (0.0024)	0.0010 (0.0013)	-0.0014 (0.0061)	0.0000 (0.0027)
Wealth (Medium)	0.1266*** (0.0364)	-0.0611*** (0.0105)	-0.0333*** (0.0074)	0.0371 (0.0241)	-0.0052 (0.0066)	-0.0022 (0.0034)	0.1081*** (0.0257)	-0.0483*** (0.0102)
Wealth (High)	0.4123*** (0.0339)	-0.1515*** (0.0135)	-0.0679*** (0.0081)	0.1709*** (0.0247)	-0.0221*** (0.0067)	-0.0037 (0.0032)	0.3622*** (0.0225)	-0.1228*** (0.0088)
Urban	0.0853** (0.0343)	-0.0025 (0.0098)	0.0009 (0.0067)	0.0183 (0.0291)	0.0057 (0.0079)	0.0069* (0.0036)	0.0604** (0.0298)	-0.0038 (0.0122)
Wind Speed	-0.0505 (0.0497)	0.0184 (0.0124)	0.0264** (0.0117)	-0.1826*** (0.0307)	0.0324*** (0.0079)	0.0123*** (0.0045)	-0.1512*** (0.0388)	0.0448*** (0.0135)
Rel. Humidity	-0.0333** (0.0145)	0.0117** (0.0049)	0.0118*** (0.0025)	0.0252* (0.0144)	-0.0050 (0.0040)	-0.0027 (0.0018)	-0.0040 (0.0122)	0.0037 (0.0046)
Temperature	-0.0196 (0.0248)	-0.0055 (0.0095)	-0.0027 (0.0065)	0.1050*** (0.0263)	-0.0274*** (0.0065)	-0.0047 (0.0037)	0.0665*** (0.0255)	-0.0307*** (0.0074)
Precipitation	-0.0010 (0.0006)	0.0004* (0.0002)	0.0001 (0.0001)	0.0001 (0.0006)	-0.0000 (0.0002)	0.0000 (0.0001)	-0.0005 (0.0005)	0.0002 (0.0002)
Observations	19,277	19,277	19,277	19,277	19,277	19,277	19,277	19,277
Cluster FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Full-control version of the age heterogeneity panel. Standard errors clustered at the thermal-inversion grid level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B5: Heterogeneity: Birth Order, Full Coefficients

	HAZ	Stunt.	Sev. Stunt.	WHZ	Wast.	Sev. Wast.	WAZ	Underwt.
Predicted PM2.5	-0.0147** (0.0074)	0.0050** (0.0023)	0.0016 (0.0014)	0.0087* (0.0051)	-0.0035** (0.0015)	-0.0010 (0.0007)	-0.0006 (0.0050)	0.0017 (0.0021)
PM2.5 × Birth order 2	0.0002 (0.0027)	0.0007 (0.0011)	0.0016** (0.0007)	0.0030 (0.0032)	0.0008 (0.0009)	-0.0001 (0.0004)	0.0015 (0.0024)	-0.0012 (0.0011)
PM2.5 × Birth order 3	0.0019 (0.0038)	-0.0020 (0.0022)	-0.0001 (0.0011)	-0.0017 (0.0037)	0.0018* (0.0010)	0.0007 (0.0006)	-0.0012 (0.0029)	-0.0017 (0.0015)
PM2.5 × Birth order 4+	-0.0011 (0.0039)	0.0001 (0.0013)	-0.0012 (0.0015)	0.0010 (0.0035)	0.0012 (0.0009)	-0.0008 (0.0006)	-0.0005 (0.0036)	-0.0007 (0.0026)
Birth order 2	0.0331 (0.1660)	-0.0444 (0.0660)	-0.0889** (0.0442)	-0.2136 (0.1983)	-0.0588 (0.0584)	0.0039 (0.0250)	-0.0794 (0.1459)	0.0656 (0.0662)
Birth order 3	-0.0545 (0.2295)	0.1146 (0.1404)	0.0059 (0.0666)	0.0545 (0.2230)	-0.1226** (0.0601)	-0.0411 (0.0394)	0.0812 (0.1762)	0.0927 (0.0908)
Birth order 4+	0.0859 (0.2348)	0.0057 (0.0738)	0.0908 (0.0903)	-0.0942 (0.2094)	-0.0742 (0.0567)	0.0533 (0.0332)	0.0212 (0.2146)	0.0451 (0.1599)
Child Age	-0.0076*** (0.0010)	0.0012*** (0.0004)	-0.0003* (0.0002)	-0.0038*** (0.0008)	-0.0010*** (0.0002)	-0.0007*** (0.0001)	-0.0070*** (0.0007)	0.0015*** (0.0003)
Male	0.0044 (0.0177)	0.0065 (0.0078)	-0.0008 (0.0044)	-0.0179 (0.0151)	0.0130*** (0.0047)	0.0061** (0.0025)	0.0352** (0.0165)	-0.0101 (0.0066)
Mother's Educ.	0.0640*** (0.0044)	-0.0172*** (0.0013)	-0.0079*** (0.0008)	0.0218*** (0.0029)	-0.0025*** (0.0008)	-0.0005 (0.0005)	0.0530*** (0.0035)	-0.0129*** (0.0011)
Household Size	-0.0031 (0.0044)	0.0021 (0.0014)	0.0006 (0.0010)	0.0030 (0.0039)	-0.0004 (0.0011)	-0.0002 (0.0005)	-0.0006 (0.0045)	0.0003 (0.0015)
Wealth (Medium)	0.1297*** (0.0366)	-0.0624*** (0.0106)	-0.0342*** (0.0073)	0.0391 (0.0238)	-0.0059 (0.0066)	-0.0026 (0.0034)	0.1097*** (0.0257)	-0.0486*** (0.0102)
Wealth (High)	0.4176*** (0.0348)	-0.1534*** (0.0140)	-0.0692*** (0.0079)	0.1725*** (0.0240)	-0.0221*** (0.0067)	-0.0041 (0.0032)	0.3638*** (0.0229)	-0.1231*** (0.0089)
Urban	0.0990*** (0.0371)	-0.0069 (0.0102)	-0.0016 (0.0068)	0.0174 (0.0286)	0.0060 (0.0078)	0.0067* (0.0036)	0.0643** (0.0294)	-0.0051 (0.0120)
Wind Speed	-0.0375 (0.0524)	0.0142 (0.0139)	0.0244** (0.0118)	-0.1752*** (0.0305)	0.0313*** (0.0079)	0.0121*** (0.0045)	-0.1422*** (0.0388)	0.0423*** (0.0137)
Rel. Humidity	-0.0275* (0.0153)	0.0115** (0.0052)	0.0118*** (0.0024)	0.0284** (0.0144)	-0.0038 (0.0039)	-0.0021 (0.0018)	0.0030 (0.0123)	0.0020 (0.0048)
Temperature	0.0398 (0.0320)	-0.0203* (0.0111)	-0.0105 (0.0070)	0.0766*** (0.0245)	-0.0230*** (0.0067)	-0.0048 (0.0036)	0.0748*** (0.0258)	-0.0313*** (0.0087)
Precipitation	-0.0012* (0.0007)	0.0004 (0.0003)	0.0001 (0.0001)	-0.0002 (0.0006)	-0.0000 (0.0002)	-0.0000 (0.0001)	-0.0009* (0.0005)	0.0003 (0.0002)
Observations	19,277	19,277	19,277	19,277	19,277	19,277	19,277	19,277
Cluster FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Full-control version of the birth order heterogeneity panel. Standard errors clustered at the thermal-inversion grid level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B6: Heterogeneity: Urban, Full Coefficients

	HAZ	Stunt.	Sev. Stunt.	WHZ	Wast.	Sev. Wast.	WAZ	Underwt.
Predicted PM2.5	-0.0151* (0.0080)	0.0045 (0.0028)	0.0017 (0.0015)	0.0081 (0.0053)	-0.0029* (0.0015)	-0.0010 (0.0007)	-0.0015 (0.0052)	0.0015 (0.0022)
PM2.5 × Urban	0.0027 (0.0051)	0.0015 (0.0014)	0.0003 (0.0011)	0.0055 (0.0035)	0.0007 (0.0009)	-0.0001 (0.0005)	0.0042 (0.0039)	-0.0027 (0.0017)
Urban	-0.0758 (0.3227)	-0.1060 (0.0896)	-0.0219 (0.0688)	-0.3372 (0.2232)	-0.0365 (0.0550)	0.0154 (0.0313)	-0.2044 (0.2452)	0.1678 (0.1097)
Male	0.0052 (0.0175)	0.0066 (0.0077)	-0.0007 (0.0044)	-0.0176 (0.0150)	0.0129*** (0.0047)	0.0061** (0.0025)	0.0358** (0.0164)	-0.0105 (0.0065)
Child Age	-0.0077*** (0.0010)	0.0011*** (0.0004)	-0.0003* (0.0002)	-0.0038*** (0.0008)	-0.0010*** (0.0002)	-0.0007*** (0.0001)	-0.0071*** (0.0007)	0.0015*** (0.0003)
Mother's Educ.	0.0640*** (0.0044)	-0.0171*** (0.0013)	-0.0079*** (0.0008)	0.0217*** (0.0029)	-0.0023*** (0.0008)	-0.0006 (0.0005)	0.0529*** (0.0036)	-0.0128*** (0.0011)
Household Size	-0.0034 (0.0044)	0.0022 (0.0014)	0.0007 (0.0010)	0.0036 (0.0039)	-0.0004 (0.0011)	-0.0002 (0.0005)	-0.0005 (0.0045)	0.0003 (0.0015)
Birth Order	0.0111 (0.0075)	0.0020 (0.0028)	0.0031* (0.0018)	-0.0130** (0.0062)	-0.0004 (0.0024)	0.0011 (0.0013)	-0.0015 (0.0059)	0.0001 (0.0027)
Wealth (Medium)	0.1311*** (0.0365)	-0.0622*** (0.0106)	-0.0340*** (0.0074)	0.0398* (0.0239)	-0.0058 (0.0066)	-0.0026 (0.0034)	0.1108*** (0.0256)	-0.0493*** (0.0102)
Wealth (High)	0.4181*** (0.0350)	-0.1545*** (0.0141)	-0.0692*** (0.0080)	0.1689*** (0.0241)	-0.0230*** (0.0065)	-0.0039 (0.0033)	0.3625*** (0.0231)	-0.1221*** (0.0089)
Wind Speed	-0.0409 (0.0529)	0.0128 (0.0143)	0.0244** (0.0118)	-0.1815*** (0.0298)	0.0302*** (0.0078)	0.0122** (0.0048)	-0.1468*** (0.0392)	0.0459*** (0.0135)
Rel. Humidity	-0.0274* (0.0151)	0.0117** (0.0051)	0.0120*** (0.0024)	0.0292** (0.0145)	-0.0037 (0.0039)	-0.0020 (0.0018)	0.0035 (0.0124)	0.0018 (0.0048)
Temperature	0.0429 (0.0331)	-0.0197* (0.0113)	-0.0113 (0.0072)	0.0800*** (0.0245)	-0.0217*** (0.0066)	-0.0050 (0.0036)	0.0775*** (0.0270)	-0.0342*** (0.0090)
Precipitation	-0.0012* (0.0007)	0.0004 (0.0003)	0.0001 (0.0001)	-0.0002 (0.0006)	-0.0000 (0.0002)	-0.0000 (0.0001)	-0.0009* (0.0005)	0.0003 (0.0002)
Observations	19,277	19,277	19,277	19,277	19,277	19,277	19,277	19,277
Cluster FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Full-control version of the urban heterogeneity panel. Standard errors clustered at the thermal-inversion grid level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B7: Flexible Weather Controls, Full Coefficients

	HAZ	Stunt.	Sev. Stunt.	WHZ	Wast.	Sev. Wast.	WAZ	Underwt.
Predicted PM2.5	-0.0154*	0.0058**	0.0037**	0.0147***	-0.0032**	-0.0013*	0.0027	0.0017
	(0.0084)	(0.0029)	(0.0019)	(0.0049)	(0.0013)	(0.0007)	(0.0054)	(0.0025)
Male	0.0051	0.0068	-0.0004	-0.0200	0.0133***	0.0063**	0.0342**	-0.0096
	(0.0177)	(0.0078)	(0.0045)	(0.0147)	(0.0048)	(0.0025)	(0.0165)	(0.0066)
Child age (months)	-0.0076***	0.0011***	-0.0003	-0.0036***	-0.0010***	-0.0007***	-0.0070***	0.0013***
	(0.0010)	(0.0004)	(0.0002)	(0.0008)	(0.0002)	(0.0001)	(0.0007)	(0.0003)
Mother's education (years)	0.0643***	-0.0171***	-0.0078***	0.0218***	-0.0022***	-0.0006	0.0531***	-0.0127***
	(0.0045)	(0.0013)	(0.0008)	(0.0029)	(0.0009)	(0.0005)	(0.0036)	(0.0011)
Household size	-0.0030	0.0021	0.0005	0.0030	-0.0004	-0.0001	-0.0007	0.0003
	(0.0045)	(0.0014)	(0.0010)	(0.0038)	(0.0011)	(0.0005)	(0.0043)	(0.0015)
Birth order	0.0125	0.0017	0.0029	-0.0123**	-0.0004	0.0010	-0.0002	-0.0002
	(0.0077)	(0.0028)	(0.0018)	(0.0061)	(0.0024)	(0.0013)	(0.0060)	(0.0027)
Wealth: medium	0.1276***	-0.0619***	-0.0350***	0.0468**	-0.0082	-0.0038	0.1145***	-0.0503***
	(0.0359)	(0.0105)	(0.0074)	(0.0231)	(0.0064)	(0.0033)	(0.0249)	(0.0101)
Wealth: high	0.4125***	-0.1532***	-0.0711***	0.1821***	-0.0263***	-0.0060*	0.3680***	-0.1260***
	(0.0336)	(0.0131)	(0.0079)	(0.0234)	(0.0064)	(0.0031)	(0.0208)	(0.0087)
Urban	0.0927***	-0.0047	-0.0019	-0.0004	0.0096	0.0074**	0.0462	-0.0027
	(0.0353)	(0.0103)	(0.0072)	(0.0288)	(0.0078)	(0.0037)	(0.0289)	(0.0117)
Weather controls (deciles)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	19,277	19,277	19,277	19,277	19,277	19,277	19,277	19,277

All control coefficients reported across the eight child-growth outcomes. Wind speed, relative humidity, temperature, and precipitation each enter as decile indicators (nine dummies per variable); their coefficients are suppressed for brevity. Standard errors clustered at the thermal-inversion grid level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B8: Excluding Weather Controls, Full Coefficients

	HAZ	Stunt.	Sev. Stunt.	WHZ	Wast.	Sev. Wast.	WAZ	Underwt.
Predicted PM2.5	-0.0029	0.0006	-0.0002	0.0054**	-0.0015**	-0.0006*	0.0037	-0.0009
	(0.0039)	(0.0013)	(0.0009)	(0.0027)	(0.0007)	(0.0003)	(0.0031)	(0.0011)
Male	0.0062	0.0060	-0.0010	-0.0180	0.0130***	0.0061**	0.0362**	-0.0105
	(0.0176)	(0.0078)	(0.0044)	(0.0151)	(0.0048)	(0.0025)	(0.0166)	(0.0065)
Child age (months)	-0.0089***	0.0017***	0.0001	-0.0033***	-0.0010***	-0.0007***	-0.0075***	0.0018***
	(0.0010)	(0.0004)	(0.0002)	(0.0008)	(0.0002)	(0.0001)	(0.0006)	(0.0002)
Mother's education (years)	0.0659***	-0.0178***	-0.0084***	0.0215***	-0.0022***	-0.0005	0.0539***	-0.0132***
	(0.0043)	(0.0013)	(0.0008)	(0.0030)	(0.0008)	(0.0005)	(0.0036)	(0.0011)
Household size	-0.0049	0.0027*	0.0009	0.0041	-0.0005	-0.0002	-0.0011	0.0006
	(0.0045)	(0.0015)	(0.0010)	(0.0039)	(0.0011)	(0.0005)	(0.0045)	(0.0014)
Birth order	0.0099	0.0027	0.0033*	-0.0117*	-0.0005	0.0010	-0.0013	0.0002
	(0.0077)	(0.0029)	(0.0019)	(0.0063)	(0.0024)	(0.0013)	(0.0059)	(0.0027)
Wealth: medium	0.1303***	-0.0628***	-0.0346***	0.0470**	-0.0081	-0.0031	0.1153***	-0.0508***
	(0.0356)	(0.0107)	(0.0072)	(0.0235)	(0.0067)	(0.0033)	(0.0250)	(0.0100)
Wealth: high	0.4058***	-0.1490***	-0.0670***	0.1847***	-0.0260***	-0.0050	0.3652***	-0.1231***
	(0.0321)	(0.0130)	(0.0079)	(0.0240)	(0.0066)	(0.0031)	(0.0210)	(0.0084)
Urban	0.0693**	0.0042	0.0084	0.0139	0.0049	0.0069**	0.0439	0.0021
	(0.0312)	(0.0084)	(0.0080)	(0.0257)	(0.0065)	(0.0030)	(0.0274)	(0.0099)
Wind speed (m/s)	—	—	—	—	—	—	—	—
Relative humidity (%)	—	—	—	—	—	—	—	—
Temperature (C)	—	—	—	—	—	—	—	—
Precipitation (mm)	—	—	—	—	—	—	—	—
Observations	19,277	19,277	19,277	19,277	19,277	19,277	19,277	19,277

All control coefficients reported across the eight child-growth outcomes. Standard errors clustered at the thermal-inversion grid level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B9: Winsorized Outcomes, Full Coefficients

	HAZ	Stunt.	Sev. Stunt.	WHZ	Wast.	Sev. Wast.	WAZ	Underwt.
Predicted PM2.5	-0.0151*	0.0048*	0.0018	0.0087*	-0.0028*	-0.0010	-0.0008	0.0009
	(0.0079)	(0.0027)	(0.0015)	(0.0048)	(0.0015)	(0.0007)	(0.0051)	(0.0022)
Male	0.0056	0.0064	-0.0008	-0.0193	0.0129***	0.0061**	0.0311*	-0.0103
	(0.0179)	(0.0078)	(0.0044)	(0.0145)	(0.0047)	(0.0025)	(0.0163)	(0.0066)
Child age (months)	-0.0073***	0.0012***	-0.0003*	-0.0038***	-0.0010***	-0.0007***	-0.0071***	0.0015***
	(0.0010)	(0.0004)	(0.0002)	(0.0008)	(0.0002)	(0.0001)	(0.0007)	(0.0003)
Mother's education (years)	0.0624***	-0.0171***	-0.0080***	0.0202***	-0.0023***	-0.0006	0.0504***	-0.0128***
	(0.0042)	(0.0013)	(0.0008)	(0.0027)	(0.0008)	(0.0005)	(0.0034)	(0.0011)
Household size	-0.0039	0.0021	0.0007	0.0039	-0.0004	-0.0002	-0.0005	0.0004
	(0.0042)	(0.0014)	(0.0010)	(0.0038)	(0.0011)	(0.0005)	(0.0043)	(0.0014)
Birth order	0.0103	0.0021	0.0032*	-0.0139**	-0.0004	0.0011	-0.0016	0.0000
	(0.0074)	(0.0028)	(0.0018)	(0.0059)	(0.0024)	(0.0013)	(0.0058)	(0.0027)
Wealth: medium	0.1311***	-0.0625***	-0.0341***	0.0405*	-0.0060	-0.0025	0.1103***	-0.0488***
	(0.0352)	(0.0106)	(0.0074)	(0.0231)	(0.0066)	(0.0034)	(0.0254)	(0.0102)
Wealth: high	0.4115***	-0.1538***	-0.0691***	0.1675***	-0.0227***	-0.0040	0.3607***	-0.1234***
	(0.0346)	(0.0140)	(0.0080)	(0.0234)	(0.0066)	(0.0032)	(0.0227)	(0.0089)
Urban	0.0908***	-0.0064	-0.0008	0.0204	0.0057	0.0069*	0.0593**	-0.0049
	(0.0348)	(0.0102)	(0.0068)	(0.0280)	(0.0079)	(0.0036)	(0.0288)	(0.0121)
Wind speed (m/s)	-0.0298	0.0148	0.0248**	-0.1678***	0.0310***	0.0120***	-0.1318***	0.0425***
	(0.0520)	(0.0139)	(0.0118)	(0.0296)	(0.0078)	(0.0045)	(0.0381)	(0.0139)
Relative humidity (%)	-0.0279*	0.0116**	0.0120***	0.0298**	-0.0038	-0.0020	0.0033	0.0021
	(0.0150)	(0.0051)	(0.0024)	(0.0141)	(0.0039)	(0.0018)	(0.0119)	(0.0048)
Temperature (C)	0.0404	-0.0212*	-0.0116	0.0695***	-0.0223***	-0.0049	0.0700***	-0.0316***
	(0.0324)	(0.0112)	(0.0073)	(0.0236)	(0.0066)	(0.0035)	(0.0263)	(0.0090)
Precipitation (mm)	-0.0013*	0.0004	0.0001	-0.0003	-0.0000	-0.0000	-0.0009*	0.0003
	(0.0007)	(0.0003)	(0.0001)	(0.0006)	(0.0002)	(0.0001)	(0.0005)	(0.0002)
Observations	19,277	19,277	19,277	19,277	19,277	19,277	19,277	19,277

All control coefficients reported across the eight child-growth outcomes. Standard errors clustered at the thermal-inversion grid level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B10: Gestational Window (Conception to Birth), Full Coefficients

	HAZ	Stunt.	Sev. Stunt.	WHZ	Wast.	Sev. Wast.	WAZ	Underwt.
Predicted PM2.5	-0.0036*** (0.0011)	0.0009** (0.0004)	0.0006** (0.0003)	-0.0007 (0.0010)	-0.0001 (0.0003)	-0.0001 (0.0001)	-0.0020** (0.0009)	0.0006 (0.0004)
Male	0.0067 (0.0172)	0.0058 (0.0077)	-0.0010 (0.0043)	-0.0194 (0.0151)	0.0132*** (0.0047)	0.0063** (0.0025)	0.0354** (0.0164)	-0.0104 (0.0065)
Child age (months)	-0.0081*** (0.0011)	0.0013*** (0.0004)	-0.0003 (0.0002)	-0.0035*** (0.0008)	-0.0010*** (0.0002)	-0.0007*** (0.0001)	-0.0071*** (0.0007)	0.0015*** (0.0003)
Mother's education (years)	0.0646*** (0.0043)	-0.0173*** (0.0013)	-0.0080*** (0.0008)	0.0210*** (0.0030)	-0.0021** (0.0009)	-0.0005 (0.0005)	0.0527*** (0.0036)	-0.0128*** (0.0011)
Household size	-0.0040 (0.0045)	0.0023 (0.0015)	0.0008 (0.0010)	0.0039 (0.0039)	-0.0005 (0.0011)	-0.0002 (0.0005)	-0.0006 (0.0045)	0.0004 (0.0014)
Birth order	0.0114 (0.0075)	0.0020 (0.0028)	0.0031* (0.0018)	-0.0132** (0.0062)	-0.0003 (0.0024)	0.0011 (0.0013)	-0.0015 (0.0059)	0.0000 (0.0028)
Wealth: medium	0.1230*** (0.0356)	-0.0597*** (0.0104)	-0.0332*** (0.0074)	0.0456* (0.0236)	-0.0078 (0.0065)	-0.0031 (0.0033)	0.1108*** (0.0253)	-0.0486*** (0.0101)
Wealth: high	0.4056*** (0.0321)	-0.1485*** (0.0127)	-0.0676*** (0.0080)	0.1865*** (0.0246)	-0.0265*** (0.0064)	-0.0051* (0.0030)	0.3676*** (0.0215)	-0.1232*** (0.0083)
Urban	0.0751** (0.0337)	0.0019 (0.0095)	0.0021 (0.0060)	0.0372 (0.0250)	0.0006 (0.0067)	0.0051 (0.0031)	0.0660** (0.0290)	-0.0039 (0.0119)
Wind speed (m/s)	-0.0855** (0.0367)	0.0317*** (0.0097)	0.0305*** (0.0089)	-0.1350*** (0.0248)	0.0203*** (0.0057)	0.0084*** (0.0032)	-0.1384*** (0.0299)	0.0445*** (0.0100)
Relative humidity (%)	-0.0302** (0.0147)	0.0128** (0.0051)	0.0122*** (0.0025)	0.0335** (0.0131)	-0.0049 (0.0036)	-0.0023 (0.0018)	0.0050 (0.0122)	0.0018 (0.0049)
Temperature (C)	0.0591** (0.0237)	-0.0300*** (0.0082)	-0.0132** (0.0061)	0.0392** (0.0191)	-0.0143*** (0.0049)	-0.0027 (0.0031)	0.0603*** (0.0184)	-0.0299*** (0.0063)
Precipitation (mm)	-0.0006 (0.0004)	0.0002 (0.0002)	0.0000 (0.0001)	-0.0011*** (0.0004)	0.0002** (0.0001)	0.0001 (0.0000)	-0.0011*** (0.0004)	0.0003** (0.0001)
Observations	19,277	19,277	19,277	19,277	19,277	19,277	19,277	19,277

All control coefficients reported across the eight child-growth outcomes. Standard errors clustered at the thermal-inversion grid level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B11: First 1,000 Days (Conception to 24 Months), Full Coefficients

	HAZ	Stunt.	Sev. Stunt.	WHZ	Wast.	Sev. Wast.	WAZ	Underwt.
Predicted PM2.5	-0.0092 (0.0075)	0.0044* (0.0025)	0.0021 (0.0019)	-0.0031 (0.0059)	-0.0019 (0.0016)	-0.0010 (0.0007)	-0.0083 (0.0059)	0.0019 (0.0026)
Male	0.0820*** (0.0268)	-0.0104 (0.0104)	-0.0149** (0.0062)	0.0099 (0.0217)	0.0075 (0.0061)	0.0016 (0.0027)	0.1124*** (0.0219)	-0.0324*** (0.0087)
Child age (months)	0.0029 (0.0021)	-0.0015* (0.0008)	-0.0007 (0.0005)	-0.0041** (0.0019)	0.0006 (0.0005)	-0.0004* (0.0002)	-0.0029* (0.0016)	0.0008 (0.0008)
Mother's education (years)	0.0723*** (0.0046)	-0.0189*** (0.0017)	-0.0094*** (0.0009)	0.0180*** (0.0041)	-0.0014 (0.0011)	-0.0001 (0.0006)	0.0554*** (0.0046)	-0.0128*** (0.0017)
Household size	-0.0058 (0.0044)	0.0030* (0.0017)	0.0020 (0.0012)	0.0046 (0.0049)	-0.0011 (0.0013)	-0.0005 (0.0005)	-0.0007 (0.0046)	0.0012 (0.0016)
Birth order	0.0011 (0.0083)	0.0078** (0.0038)	-0.0044* (0.0026)	-0.0099 (0.0084)	0.0008 (0.0033)	0.0010 (0.0015)	-0.0063 (0.0071)	0.0024 (0.0044)
Wealth: medium	0.1515*** (0.0348)	-0.0833*** (0.0127)	-0.0382*** (0.0096)	-0.0088 (0.0279)	0.0028 (0.0079)	0.0052 (0.0041)	0.0855*** (0.0284)	-0.0543*** (0.0133)
Wealth: high	0.4806*** (0.0395)	-0.1931*** (0.0183)	-0.0768*** (0.0113)	0.1421*** (0.0298)	-0.0139 (0.0093)	-0.0026 (0.0041)	0.3840*** (0.0261)	-0.1378*** (0.0124)
Urban	0.1162*** (0.0357)	-0.0131 (0.0149)	-0.0064 (0.0077)	0.0487 (0.0321)	-0.0040 (0.0099)	0.0084* (0.0047)	0.1024*** (0.0343)	-0.0130 (0.0165)
Wind speed (m/s)	0.0046 (0.0551)	-0.0093 (0.0165)	0.0254 (0.0180)	-0.0988* (0.0506)	0.0178* (0.0098)	0.0031 (0.0044)	-0.0625 (0.0453)	0.0257 (0.0168)
Relative humidity (%)	-0.0262* (0.0147)	0.0110* (0.0063)	0.0129*** (0.0039)	0.0060 (0.0188)	0.0042 (0.0057)	0.0017 (0.0021)	-0.0117 (0.0153)	0.0069 (0.0070)
Temperature (C)	0.0425 (0.0363)	-0.0214* (0.0116)	0.0018 (0.0122)	0.0343 (0.0322)	-0.0184* (0.0108)	0.0042 (0.0045)	0.0539* (0.0314)	-0.0274** (0.0116)
Precipitation (mm)	-0.0011 (0.0008)	0.0004 (0.0003)	0.0002 (0.0002)	-0.0000 (0.0008)	-0.0003 (0.0002)	-0.0000 (0.0001)	-0.0007 (0.0007)	0.0002 (0.0003)
Observations	11,359	11,359	11,359	11,359	11,359	11,359	11,359	11,359

All control coefficients reported across the eight child-growth outcomes. Standard errors clustered at the thermal-inversion grid level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.